
When Does Pooling Pay? A Credibility Diagnostic for Forgetting and Pooling in Intermittent-Demand Forecasting

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Abstract

Forecasting many sparse series forces two coupled choices: how much of each series’ own past to keep, and how much to borrow from other series. We show that in a hierarchical empirical-Bayes hurdle model with one exponential recency operator applied to item- and group-level statistics alike, the two are one decision. Forgetting keeps a shared prior’s leverage alive while eroding the information that resolves fine group structure; a two-sided credibility bound makes this precise, and its resolution corollary turns it into a diagnostic computable from the fitting window, before any pooling structure is fitted. On five public intermittent-demand panels (more than 19,000 series), read against a controlled surface over leverage and group separation, the diagnostic is right in both directions. Where credibility is low the shared prior pays: the advantage over a per-series Gaussian-process specialist grows from two to three percent at full length to seven to eleven percent as histories shrink. Where credibility is high, refining the partition is worth nothing and nothing is found; on the one panel with room the mixture objective fails to collect it, in data and in simulation alike. The forecaster that carries the diagnostic, EBB, selects pooling structure, forgetting rate, and calibration layer on one validation tail, and is, together with that specialist, the only method within five percent of the best competitor on every panel under both fixed-origin and strict walk-forward protocols, at one to three orders of magnitude lower cost and with closed-form online updates. Two evaluation conventions

obscure such results: absolute-error metrics reward the all-zero forecast on three panels, and static split-conformal wrappers understate what conformal prediction achieves by about twenty percent.

1 Introduction

Whether to fit each series on its own or to pool evidence across series is the oldest question in forecasting many series at once, and the working answer is still empirical: global models win on some collections and lose on others, and one finds out by running both [16, 28, 29]. The question is sharpest for intermittent demand—long runs of zeros interrupted by sporadic positive quantities, pervasive in spare parts, long-tail commerce, and supply chains—because each series carries the least evidence of its own. There, a second choice compounds the first. Old observations can be discounted to track drift and obsolescence, but forgetting also removes information. Local methods such as Croston, SBA, and TSB [5, 23, 25] adapt in time while estimating each item separately; hierarchical probabilistic models [2, 18, 20] shrink across items, usually under a pooling structure and temporal model fixed in advance. Pooling and forgetting are consequently designed as separate decisions.

They are not separate statistically. With exponential forgetting, an item’s effective history remains bounded, so a shared prior can retain substantial influence even after a long observation sequence. The same discounting, however, reduces the effective information used to estimate each group-level prior. Stronger forgetting can therefore make pooling more consequential while making fine pooling structure harder to resolve. This tension is the paper’s organizing principle: the forgetting rate sets not only a temporal horizon but also the statistical resolution at which cross-series structure can be learned, and both sides of the trade-off are functions of one quantity—the credibility weight an item retains after discounting—that can be computed from the fitting window before any partition is fitted.

We develop this principle inside EBB,¹ a recency-weighted hierarchical empirical-Bayes forecaster with Beta–Binomial occurrence and log-normal size blocks represented by per-item sufficient statistics. The same exponential weights enter the item posterior and the group-level hyperparameters. A small candidate family crosses discount factors with a global pool, the ADI/CV² taxonomy, and BIC-selected mixture partitions; one chronological validation tail selects the discount, the structure, and, under deployment, whether an adaptive conformal layer is applied. Estimation is closed form apart from the mixture search, deterministic, and linear in the number of observations.

The theory predicts an asymmetric pattern, and the experiments confirm both halves of it. Where credibility is low the shared prior should carry the forecast: truncating histories on five public panels, EBB’s advantage over the strongest per-series comparator, an official Tweedie–Gaussian-process implementation, grows from two to three percent at full length to seven to eleven percent with six months of history. Where credibility is high, refining the pooling structure should be worth little: a controlled surface that varies leverage and group separation together places four panels where a supplied true partition would earn nothing, and nothing is found; it places Carparts where one would earn 1.7 to 2.0%, and the learned partition returns −0.08% there, as the same objective does at those coordinates in simulation; a screen built from the single-pool fit alone identifies, in simulation, the regimes where refinement cannot pay materially. Forgetting itself is a first-order decision: all five panels select it, and where both effects are measurable it is worth an order of magnitude more than refinement.

The forecaster is competitive on the terms that matter for deployment. Across the five panels and both a fixed-origin and a strict walk-forward protocol, EBB and the Gaussian-process specialist are the only methods within five percent of the best competitor everywhere; EBB leads on two panels, trails by under one percent on two, and by 2.9% on M5 at fixed origin, at one to three orders of magnitude lower cost and with online updates that need no refitting.

Contributions. (1) **A two-sided resolution theory and an ex ante diagnostic.** Proposition 1 shows that one recency operator keeps prior leverage active but raises the sampling-variance floor against which group heterogeneity must be resolved; Corollary 1 converts this into a statistic computable from the fitting window that bounds what any partition

could be worth. A controlled surface over leverage and separation calibrates the bound, a pre-fit screen built from the single-pool fit identifies regimes without material room, and five real panels confirm the diagnostic in both directions (Section 5.2). (2) **A forecaster that carries the diagnostic.** EBB selects pooling structure, forgetting rate, and calibration layer on one leakage-safe validation tail; its predictive law is closed form, bit-reproducible, costs milliseconds per series, and updates online. It matches a per-series Gaussian-process specialist to within a percent on most panels and is the only other method within five percent of the best competitor on all of them (Section 5.3). (3) **Two evaluation cautions.** Absolute-error metrics reward the all-zero forecast on three of five panels and reverse the ranking of a known-correct partition in simulation, and static split-conformal wrappers understate what conformal prediction achieves by about twenty percent relative to an adaptive construction (Section 5.4). Comparators include the official TweedieGP implementation, adaptive conformal inference applied to baselines and to EBB alike, a published hierarchical state-space model, and a zero-shot foundation model.

2 Related Work

Intermittent-demand forecasting. Croston’s decomposition [5], its SBA correction [23], and TSB [25] estimate occurrence and positive size locally; TSB additionally lets the occurrence estimate decay under obsolescence. ADIDA [17] and IMAPA [12] instead reduce intermittency through temporal aggregation, while likelihood-based alternatives include inventory count models [21], iETS [22], Tweedie Gaussian processes [6], and deep renewal processes [27]. Conformal wrappers turn point forecasters into predictive intervals, and adaptive conformal inference updates miscoverage levels under distribution shift [7]; we evaluate both the static and the adaptive form, because the difference between them is larger than most method differences reported on these panels. These approaches provide important forms of temporal adaptation, but do not jointly learn how recency-weighted evidence should be shared across items. Hierarchical models do share evidence: Chapados [2] and Seeger et al. [20] pool related count series at scale, and Pitkin et al. [18] develops a hierarchical hurdle model for retail. Their pooling structure or temporal evolution is generally specified in advance. EBB instead crosses recency and pooling choices within one occurrence–size empirical-Bayes model, and asks when that crossing can pay.

Learning cross-series structure. The continuum between local and global forecasting is formalized by Montero-Manso and Hyndman [16]. More re-

¹Empirical Bayes with Bounded memory: under the recency operator an item’s effective history stays bounded (Section 4), so evidence ebbs rather than accumulates.

cent work learns richer forms of cross-series specialization. For hierarchical forecasting, Cini et al. [4] learns a graph-pooled aggregation hierarchy aligned with the forecasting objective. Moirai-MoE [14] replaces human-defined frequency groups with token-level sparse-expert routing in a time-series foundation model. These approaches motivate data-adaptive structure, but address different objects: aggregation coherence or latent expert specialization. Our groups are explicit exchangeability pools for a shared prior. That restriction is what makes their statistical resolution analyzable: the quantity governing whether a pool can be estimated is the same credibility weight that governs whether it can matter, so a single diagnostic covers both. It also permits a direct comparison among global, fixed-rule demand-pattern, and learned mixture pools under one predictive law.

Discounting, credibility, and selection. Exponential discounting is classical in Bayesian dynamic models [26], and power priors down-weight historical likelihoods [9]. Likewise, $\lambda_i = n_i/(n_i + \phi_g)$ is the classical actuarial credibility form [1]; Proposition 5 is therefore a restatement, not a claimed contribution. The distinction in EBB is that one recency operator updates both item statistics and the empirical-Bayes statistics defining the shared prior, and its strength is selected jointly with the pool. Proposition 1 then identifies the resulting two-sided resolution limit: forgetting preserves prior leverage but can erase the information needed to distinguish fine pools. Chronological validation selects among the finite candidate family. Because zero-heavy point losses can favor degenerate forecasts, selection uses pinball loss, a proper scoring rule [8], with the high quantiles relevant to inventory decisions [11].

3 EBB: Recency-Weighted Hierarchical EB

EBB combines two candidate choices inside one occurrence-size model: a recency discount w determines which observations remain informative, and a partition g determines which items share an empirical-Bayes prior. The key construction is that the *same* discounted sufficient statistics determine both the item posterior and the group prior. We first define that representation, then the candidate partitions and their leakage-safe joint selection.

3.1 Occurrence-size model

For item i , let $z_{i,t} = \mathbf{1}\{y_{i,t} > 0\}$ and define $\ell_{i,t} = \log y_{i,t}$ when $z_{i,t} = 1$. Conditional on membership

$g_i = g(i)$, the hurdle model is

$$\begin{aligned} \pi_i &\sim \text{Beta}(\alpha_{g_i}, \beta_{g_i}), \quad z_{i,t} \mid \pi_i \sim \text{Bern}(\pi_i), \\ \mu_i &\sim \mathcal{N}(\mu_{0,g_i}, \tau_{g_i}^2), \quad \sigma_i^2 \sim \text{ScaleInv-}\chi^2(\nu, s_{0,g_i}^2), \\ \ell_{i,t} \mid z_{i,t}=1, \mu_i, \sigma_i^2 &\sim \mathcal{N}(\mu_i, \sigma_i^2). \end{aligned} \quad (1)$$

Here $s_{0,g}^2$ is parameterized so that the prior mean of σ_i^2 is the group estimate σ_g^2 ; ν is fixed rather than selected. Group hyperparameters are estimated by empirical Bayes: Beta-Binomial marginal likelihood for occurrence, and an intercept-only random-effects fit plus conjugate variance shrinkage for positive log-size. Appendix D gives the numerical objectives and fallback rules.

Without forgetting, all inference depends on the four statistics

$$S_i(1) = (n_i, c_i, u_i, v_i) = \sum_{t=1}^{T_i} (1, z_{i,t}, z_{i,t}\ell_{i,t}, z_{i,t}\ell_{i,t}^2),$$

where the last two terms are zero when $z_{i,t} = 0$. The occurrence posterior mean and the positive-size posterior location take parallel credibility forms,

$$\begin{aligned} \hat{\pi}_i &= \lambda_i^{(o)} \bar{p}_i + (1 - \lambda_i^{(o)}) \mu_{\pi, g_i}, \quad \lambda_i^{(o)} = \frac{n_i}{n_i + \phi_{g_i}}, \\ \hat{\mu}_i &= \lambda_i^{(+)} \bar{\ell}_i + (1 - \lambda_i^{(+)}) \mu_{0, g_i}, \quad \lambda_i^{(+)} = \frac{n_i^+}{n_i^+ + \kappa_{i, g_i}}, \end{aligned} \quad (2)$$

with $\phi_g = \alpha_g + \beta_g$ and $\kappa_{i, g} = \hat{\sigma}_i^2 / \tau_g^2$. Thus a partition matters only through the prior target and strength supplied to its members.

Two variance ratios appear in what follows. Estimation uses the item-level $\kappa_{i, g} = \hat{\sigma}_i^2 / \tau_g^2$ in (2); the analysis in Section 4.2 works with the group-level surrogate $\kappa_g = \sigma_g^2 / \tau_g^2$, which replaces each item's process variance by its group estimate. The two coincide when the group is homogeneous in σ_i^2 , and the surrogate is what the collapse event of Proposition 1 is stated in terms of.

The plug-in marginal predictive distribution is zero-inflated log-normal:

$$\begin{aligned} \Pr(Y_i = 0 \mid \mathcal{D}) &= 1 - \hat{\pi}_i, \\ \log Y_i \mid (Y_i > 0, \mathcal{D}) &\sim \mathcal{N}(\hat{\mu}_i, \hat{\sigma}_i^2 + V_i), \end{aligned} \quad (3)$$

where V_i is posterior uncertainty in μ_i . Its quantiles are analytic. For point evaluation we retain the conditional plug-in mean $\hat{y}_i = \hat{\pi}_i \exp(\hat{\mu}_i + \hat{\sigma}_i^2/2)$; it is distinct from the mean of (3), which also integrates V_i .

3.2 Shared recency weighting

For $w \in (0, 1]$, replace $S_i(1)$ by

$$S_i(w) = \sum_{t=1}^{T_i} w^{T_i-t} (1, z_{i,t}, z_{i,t} \ell_{i,t}, z_{i,t} \ell_{i,t}^2). \quad (4)$$

These fractional counts are valid weighted-likelihood statistics; the log-Gamma form of the occurrence marginal and the Normal size objective extend directly to them. Crucially, $S_i(w)$ is used twice:

$$\begin{aligned} \hat{\eta}_g(w) &= \arg \max_{\eta_g} \sum_{i: g_i=g} \log m\{S_i(w) \mid \eta_g\}, \\ q_i^{(w)}(\theta_i) &= q_i(\theta_i \mid S_i(w), \hat{\eta}_{g_i}(w)), \end{aligned} \quad (5)$$

where η_g collects the group hyperparameters and q_i is the item posterior. The group prior therefore adapts to recent cross-sectional evidence rather than remaining a stationary prior applied to discounted item data. Setting $w = 1$ recovers the stationary estimator exactly.

For occurrence, (2) becomes

$$\begin{aligned} \hat{\pi}_i(w) &= \lambda_i(w) \bar{p}_i(w) + \{1 - \lambda_i(w)\} \mu_{\pi, g_i}(w), \\ \lambda_i(w) &= \frac{n_i(w)}{n_i(w) + \phi_{g_i}(w)}. \end{aligned} \quad (6)$$

The analogous positive-size update uses discounted positive mass $n_i^+(w)$ and the group variance ratio $\kappa_{i,g}(w)$.

3.3 Candidate pooling structures

Let $\mathcal{R}$ contain three ways to construct $g(\cdot)$ from the fitting data, ordered coarse to fine: a *global* pool with one prior for the panel; the *taxonomy* of four ADI/CV² demand-pattern classes at the classical thresholds [24]; and a *learned* partition, hard assignments from a mixture of occurrence-size prior components with $K \in \{1, \dots, 6\}$ selected by BIC, fit by EM with an analytic responsibility update and a deterministic initialization (Appendix E, Table 2). The question this paper asks is not which rung is best in general but whether the discounted data support moving up the ladder at all; Section 4.2 gives the condition and Section 5.2 reports what happens when it fails. Every data-dependent label is computed from the selector’s fitting head, never its validation tail.

3.4 Leakage-safe joint selection

Each training series is split chronologically into a fitting head $\mathcal{H}_i$ (first 80%) and validation tail $\mathcal{V}_i$ (last 20%). For every $(r, w) \in \mathcal{R} \times \mathcal{W}$, with

$$\mathcal{W} = \{1, .999, .997, .995, .99, .98, .95, .90\},$$

the partition builder for r sees only $\mathcal{H} = \cup_i \mathcal{H}_i$. The resulting labels, discounted EB prior, and item posteriors are all fit on the same head. The candidate is scored by

$$\begin{aligned} \hat{R}(r, w) &= \frac{1}{N_V} \sum_{i \in \mathcal{I}_V} \frac{1}{|\mathcal{V}_i| |\mathcal{Q}|} \\ &\times \sum_{t \in \mathcal{V}_i} \sum_{q \in \mathcal{Q}} \frac{\rho_q(y_{i,t}, \hat{Q}_{i,t}^{(r,w)}(q))}{a_i}, \end{aligned} \quad (7)$$

where $\mathcal{Q} = \{.5, .75, .9\}$, ρ_q is pinball loss, and a_i is mean absolute demand on the fitting head (with a panel-level fallback for an all-zero head); $N_V = |\mathcal{I}_V|$ is the number of scored items. Times and quantiles are averaged within item before items receive uniform weight. This matches the panel-level goal and prevents long series or high-volume items from dominating selection.

The selected pair is $(\hat{r}, \hat{w}) = \arg \min_{r,w} \hat{R}(r, w)$. Its partition is then *relearned* on the complete training window at $\hat{w}$, and the EB model is refit once before external evaluation. Thus the validation tail chooses a construction rule and discount, not a permanent set of labels; external targets are never used. There are 24 outer forecast candidates.

Resolution diagnostics. Fitting each candidate already produces everything the resolution analysis of Section 4.2 needs: for every (r, w) and group g the selector records, from the fitting head only, the median credibility weights $\lambda_i^{(o)}$ and $\lambda_i^{(+)}$, the share of items with $\lambda_i^{(+)} < 0.1$, the group sizes, and the resolution statistic $\hat{z}_g$ whose nonpositive values coincide with the collapse event of Proposition 1 (Appendix E). They are computed before any external observation is seen, and Section 5.2 reads them.

Appendix D gives split edge cases and pseudocode-level implementation details.

3.5 Forecasts and computation

Selection uses the analytic plug-in quantiles in (3). The reported probabilistic forecasts also average quantile estimates across $B = 20$ cross-item bootstrap refits of the group hyperparameters. We treat this explicitly as a quantile ensemble; the average of component quantiles is not asserted to be the quantile of a single posterior mixture.

For each w , one pass computes $S_i(w)$ in $O(\sum_i T_i)$ time. Each EB fit requires only low-dimensional deterministic optimization per group; mixture EM is $O(NK)$ per iteration. Prediction is $O(N)$ for points and $O(NQ)$ for Q quantiles.

4 Resolution under Forgetting

The shared operator in (5) creates two opposing effects. A bounded item history prevents the group prior from vanishing, but the same bound limits the information used to estimate fine group structure. This section formalizes that interaction. Supporting results connecting the model to TSB, drift tracking, finite-class selection, and classical credibility are collected in Appendix B.

4.1 Persistent prior leverage

Fix an item and suppress its subscripts. Because $n(w) = \sum_{t=1}^T w^{T-t} \leq (1-w)^{-1}$ for $w < 1$, the occurrence credibility weight cannot converge to one: $\lambda(w) \leq \{1+(1-w)\phi\}^{-1} < 1$ for every horizon, whereas $\lambda(1) \rightarrow 1$ as $T \rightarrow \infty$ (Proposition 6, Appendix E). Forgetting keeps the pooled prior relevant indefinitely; the statement is about leverage, not benefit.

4.2 Variance-component resolution

The converse problem appears in the positive-size block. Let item i in group g have discounted positive mass n_i^+ , sampling variance $s_i = \sigma_g^2/n_i^+$ for its log-size mean, and credibility $\lambda_i^{(+)} = n_i^+/(n_i^+ + \kappa_g)$ with $\kappa_g = \sigma_g^2/\tau_g^2$. The group variance is estimated by the truncated moment/REML form $\hat{\tau}_g^2 = \max(0, S_g^2 - \bar{s}_g)$, where S_g^2 is the sample variance of the m_g item means and $\bar{s}_g = m_g^{-1} \sum_{i \in g} s_i$.

Proposition 1 (Two-sided effect of forgetting). (i) For $w < 1$, $\lambda_i^{(+)} \leq \{1 + (1-w)\kappa_g\}^{-1} < 1$. (ii) For independent item means with possibly unequal s_i ,

$$\mathbb{E}[S_g^2] = \tau_g^2 + \bar{s}_g$$

holds exactly. Under the Gaussian moment approximation,

$$\Pr(\hat{\tau}_g^2 = 0) \simeq \Phi\left(-\sqrt{\frac{m_g - 1}{2}} \rho_g\right), \quad \rho_g = \frac{\tau_g^2}{\tau_g^2 + \bar{s}_g}. \quad (8)$$

Since $n_i^+ \leq (1-w)^{-1}$, $\bar{s}_g \geq (1-w)\sigma_g^2$ and therefore

$$\rho_g \leq \frac{\tau_g^2}{\tau_g^2 + (1-w)\sigma_g^2}.$$

On the collapse event the plug-in estimate has $\hat{\tau}_g^2 = 0$ and $\lambda_i^{(+)} = 0$: the item mean is replaced by its estimated group mean.

The population ratio ρ_g is not observed. For diagnostics we use

$$\hat{z}_g = \frac{S_g^2 - \bar{s}_g}{\sqrt{2/(m_g - 1) S_g^2}},$$

whose nonpositive values coincide with the plug-in collapse event.

Corollary 1 (Resolution condition). For this estimator and window, stable estimation of a group variance requires, up to a fixed probability threshold,

$$\rho_g \sqrt{m_g - 1} \gtrsim 1, \\ \text{equivalently} \quad \tau_g^2 \gtrsim \frac{\bar{s}_g}{\sqrt{m_g - 1} - 1},$$

with $\bar{s}_g \geq (1-w)\sigma_g^2$.

For similarity-based refinements, finer groups typically reduce both m_g and the within-group spread τ_g^2 , while stronger forgetting raises the sampling-variance floor. All three changes move the estimator toward collapse. This is an estimator- and window-specific resolution condition, not an information-theoretic lower bound, and it does not say which candidate wins below the threshold.

Figure 1 summarizes why pooling can become more consequential at exactly the point where a fine partition becomes harder to estimate. Section E.7 tests both sides with known group labels. Proofs and the accuracy limits of the approximation are given in Appendix C and Section 6.

5 Experiments

5.1 Experimental design

Datasets. We evaluate on two daily panels—**Online Retail** [3] (3,649 series) and a seed-controlled 5,000-series **M5** sample [15]—and the monthly intermittent suite of Damato et al. [6]: **Auto** (3,000 series), **Carparts** (2,509), and **RAF** (5,000). Together they contain more than 19,000 series and span long daily histories and short, highly sparse monthly panels. Dataset construction and holdout lengths are in Appendix F.16. Both the probabilistic and the point M5 protocols use the first two thirds of each series for initialization.

Comparators. The classical set comprises Croston, SBA, TSB, ADIDA, and IMAPA; stronger statistical comparators are AutoARIMA, AutoTheta, a per-series autoregressive Tweedie GLM, iETS, and the official TweedieGP implementation at its released defaults on all five panels. For probabilistic evaluation we use native distributions where available and chronological split-conformal wrappers (CP-*) for the point forecasters. The walk-forward studies additionally include adaptive conformal inference (ACI) applied to ADIDA and to EBB on every panel and to TSB on Online Retail; ACI-EBB reads EBB’s analytic predictive quantile at the adaptive level, with the same γ and update

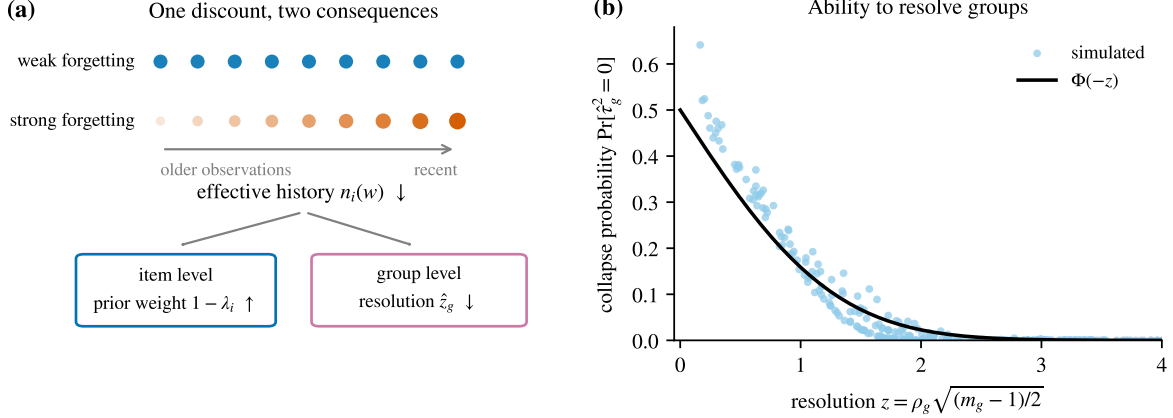


Figure 1: **One discount creates two opposing effects.** (a) The same recency operator reduces item effective history, which increases prior leverage, and group effective information, which reduces structural resolution. (b) When the partition must be estimated, the probability that the group variance collapses rises as the resolution statistic falls; the curve is the approximation in (8).

rule as ACI-ADIDA; EBB (rule) applies that layer only where a validation-tail rule inside the initialization window selects it (Section E.6). A 25-candidate tuned-TSB control receives the same internal chronological selection budget as EBB. DeepState and Chronos-Bolt comparisons are confined to Appendix F.13 because they use subset or proxy protocols. Implementation details and complete settings are in Appendix D.

Protocols and metrics. The primary fixed-origin experiment fits once on the initialization window and scores the full evaluation span. A strict walk-forward experiment then reveals and updates one block at a time: seven-day blocks on Online Retail and M5, one-month blocks on Auto, Carparts, and RAF. Classical, conformal, and TweedieGP methods refit each block, except that on Online Retail AutoARIMA, AutoTheta, iETS, and TweedieGP refit every four blocks and on M5 only the conformal and adaptive-conformal wrappers are run (Appendix F.7 gives the cost of the rest); EBB updates discounted sufficient statistics each block. The primary metric is scaled pinball loss (SPL; lower is better), averaged over $q \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$; we separately report q90 and the far tail because inventory decisions are asymmetric. All quantiles are zero-truncated and monotonized, and EBB is reported without post-hoc calibration. Every model in Table 3 is scored by one implementation from its stored quantile predictions, so no cell mixes scoring versions. Fixed-origin, walk-forward, and subset foundation model results are kept separate because they answer different questions. MAE and RMSSE are secondary diagnostics rather than the headline criteria because zero forecasts can minimize absolute error on intermittent panels. Full point, interval, significance, and efficiency

results are in Appendix F.

5.2 Where pooling pays

The diagnostic of Section 4.2 makes two predictions: that a shared prior’s return grows as an item’s own credibility falls, and that refining the pooling structure returns something only where the discounted data both leave room and resolve the structure. We test each side on the five panels, using the credibility weights at the audited configurations (Appendix E, Table 7) and a controlled study that varies leverage and group separation together (Appendix E, Table 6; Appendix F.8).

Short histories: the shared prior pays. Corollary 1 predicts where a pooled prior pays: when an item’s own evidence is thin, its credibility weight is low and the shared prior carries the forecast. Truncating the initialization window to its last L observations per series, with the evaluation span unchanged, moves every panel along that axis. Figure 2 plots EBB’s mean-SPL advantage over TweedieGP, the strongest per-series comparator, against the retained history; Table 19 gives the numbers together with the median credibility weights and the static conformal wrapper. The advantage grows as history shrinks on Auto and RAF, turns positive on Carparts only at the shortest window, stays within a third of a percent on Online Retail, and does not appear on M5, where both median credibility weights stay above 0.8 even with 28 days of history and the per-series GP leads at every length but one, by a margin that grows with the history it is given. Both methods degrade gracefully; the gap that opens at short histories is the pooled prior’s

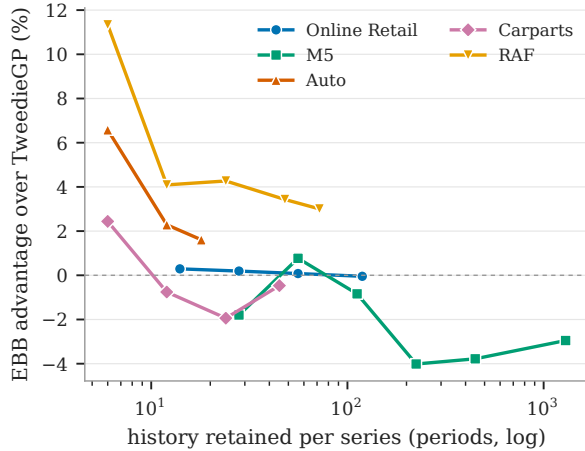


Figure 2: **Pooling pays when histories are short.** EBB’s mean-SPL advantage over TweedieGP (percent) against the history retained per series at initialization; the evaluation span is fixed and the audited structure and discount are unchanged. Full-length points reproduce Table 3.

contribution, and its size tracks the leverage the diagnostic assigns to each panel rather than the panel’s frequency.

Where the room is, and whether it is collected.

Figure 3 places the five panels on a surface that varies leverage and group separation together. Each panel’s size-block separation is measured as the share of item log-size variance explained by a partition, R^2 , under its learned partition (0.44 to 0.95) and under the objective-free ADI/CV² taxonomy (0.01 to 0.30); the simulated cells carry the same statistic under their true labels. Read within each panel’s own forgetting regime (Table 31), a supplied true partition would earn between -0.5% and $+0.6\%$ on Online Retail, M5, Auto, and RAF, and 1.7 to 2.0% on Carparts. The learned partitions return -0.05% , $+0.58\%$, -0.31% , $+0.03\%$, and -0.08% : within the surface’s range on the four panels without room, and about two points short on the one panel with it. At Carparts’s coordinates the simulated learned mixture also loses (-0.7% to -0.8%), and across the grid it is negative at every moderate separation (Figure 3, right), recovering the true partition’s return only when centers are two or more log-units apart and histories are long. The earlier reading of Online Retail as the decisive case rested on an envelope that pooled separations; resolved, its room is at most 0.6% .

This separates the two explanations that aggregate results cannot. On four panels the credibility weights and the separation together say there is little to exploit,

and the estimator’s near-zero refinement gains are the predicted outcome, not a failure. On Carparts the structure would pay and the objective does not find it; that the same objective fails on simulated panels at the same coordinates makes learnability, not the bound, the binding constraint there. Corollary 1 bounds the prize; the surface says where the prize is nonzero.

Forgetting is an order of magnitude larger.

Computed within a single selection surface under one definition—the M5 surface at its selected discount—refining the pooling structure is worth $+0.58\%$ of validation SPL, while discounting at the same structure is worth 8.95% for the global pool and 9.46% for the mixture. The asymmetry that motivates the paper survives at an order of magnitude, and it survives on the one panel where refinement is measurable at all.

A pre-fit screen. The surface answers the question ex post, with true labels; the practical question is ex ante. On an expanded controlled grid (separation, discount, length, occurrence rate, group size, imbalance; 2,700 panels), a closed-form score computed from the single-pool fit alone, $\text{med}_i(1 - \lambda_i^{(+)})^2 \hat{\tau}^2$, ranks the room a true partition could earn with held-out AUC 0.71 – 0.71 for the event “gain above one percent”; a logistic screen on the same fitting-window quantities reaches 0.84 – 0.83 when separation levels, lengths, or random folds are held out and 0.68 when a whole discount is held out, so the screen interpolates across regimes it has seen and does not extrapolate across the discount (Appendix E.9). Two facts bound the claim. 39% of the simulated single-pool fits collapse ($\hat{\tau}^2 = 0$) and half of the large oracle gains occur there: escapes from a collapsed global fit, which the resolution statistic flags, not returns to structure. And on real series with a known partition injected—group offsets of up to two log-units on Online Retail, Carparts, and M5, at the selected discount and at $w = 0.90$ —the oracle partition never earns more than 0.4% , because a single pool absorbs the injected heterogeneity into $\hat{\tau}^2$ and the items’ own evidence takes over; the screen assigns every such panel a probability below 0.17 of material room, correctly, but the test checks rejections, not detections. The screen therefore does what Corollary 1 licenses and no more: it identifies, from the fitting window, regimes where refinement cannot pay materially.

5.3 Accuracy and cost

Table 1 reports the comparison that a practitioner would make. At fixed origin, EBB attains the lowest mean SPL on Auto and RAF, by 1.6% and 3.0% over the official TweedieGP implementation, both significant; TweedieGP attains it on Online Retail, Carparts,

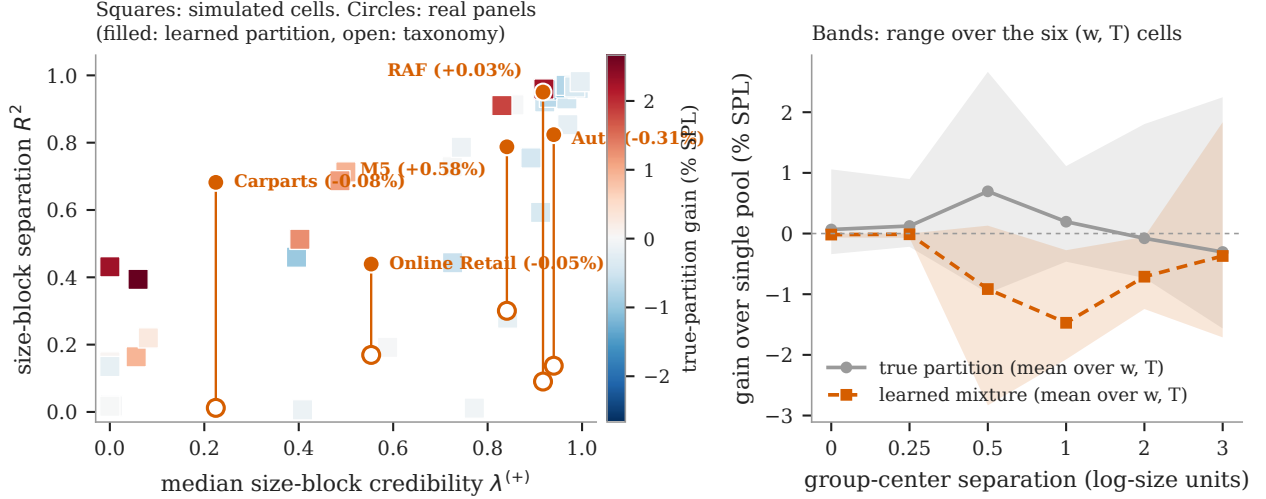


Figure 3: **Room depends on separation as well as leverage, and the real panels have little of it.** Left: each square is one cell of the controlled grid (separation $\times$ discount $\times$ length, nine paired replicates; occurrence separation scaled with size separation), placed at its median size-block credibility and at the share of item log-size variance its true partition explains (R^2), colored by the true partition’s SPL gain over a single pool. Real panels are placed at the same two coordinates, with R^2 measured under the learned partition (filled) and under the objective-free ADI/CV² taxonomy (open); labels give the learned partition’s realized refinement gain. Right: the true partition’s gain and the learned mixture’s gain against separation, averaged over the six (w, T) cells, with the range across cells shaded. Room read within each panel’s own forgetting regime is in Table 31.

and M5, by 0.05%, 0.5%, and 2.9%, of which only the M5 margin is significant. Against every other comparator EBB is best on all five panels, by 10.1% over iETS on Online Retail and 1.2% over CP-Croston on M5 (Appendix E, Table 3; significance in Appendix F, Table 23). Under strict walk-forward updating the picture repeats for the uncalibrated methods: EBB leads on Auto and RAF, TweedieGP on Online Retail and Carparts by 0.2% and 0.8%, and per-series paired tests cannot separate the two on three of the four panels where both run. Adaptive conformal inference on ADIDA leads M5 by 4.4%; the same layer applied to EBB lowers its M5 loss from 1.485 to 1.379, 2.9% below ACI-ADIDA, but degrades Online Retail and RAF, so whether to apply it is itself selected on the validation tail. The rule applies the layer on M5 and Auto, withholds it elsewhere, matches the ex-post better variant on four of five panels, and leaves EBB (rule) within 0.8% of the best competing method on every panel (Appendix E, Tables 4–5; Appendix F, Table 25).

Figure 4 summarizes both protocols by the largest shortfall to the per-panel best against measured cost. EBB and TweedieGP are the only methods within five percent of the best competitor everywhere (EBB at most 3.0% at fixed origin and 4.6% under walk-forward updating; TweedieGP 3.1% and 4.5%); every other method is at least 15% behind on some panel. The

two differ in cost by one to three orders of magnitude: TweedieGP refits per-series variational GPs at every update, so its walk-forward runs took between 36 minutes and 2.4 hours with twelve workers and project to about 51 hours on M5, whereas EBB completes each of the five walk-forward runs in under 40 seconds on one core, updating sufficient statistics in closed form (Appendix F.7). Point forecasts are competitive but not uniformly best (Appendix F.1).

5.4 Two evaluation cautions

The first caution concerns the probabilistic baseline: *the choice of conformal construction matters more here than the choice of forecaster*: making the wrapper adaptive [7] improves ADIDA’s mean SPL from 1.1576 to 0.9108, a 21% gain that is more than three times the remaining gap to EBB. Static split-conformal wrappers are the standard probabilistic baseline in this literature; on this evidence they understate what conformal prediction can do on intermittent panels, and comparisons drawn against them—including earlier versions of our own—are correspondingly generous. The second caution concerns point metrics. The all-zero forecast attains the lowest MAE on three of five panels and, in the controlled study, a known-correct partition is reported as harmful by MAE while improving every proper scoring rule (Appendix E, Table 6); absolute-

Table 1: **Mean scaled pinball loss on five panels under two protocols.** Fixed origin fits once and scores the full evaluation span; walk-forward reveals one block at a time (seven days on the daily panels, one month on the monthly ones), with every baseline refit per block and EBB updating sufficient statistics. The first row is the best of ten classical, automatic, and static split-conformal baselines in each cell (Online Retail: iETS / AutoTheta; M5: CP-Croston / CP-IMAPA; Auto: AutoARIMA / AutoARIMA; Carparts: CP-IMAPA / CP-IMAPA; RAF: CP-ADIDA / CP-ADIDA). ACI-ADIDA is adaptive conformal inference on ADIDA; EBB (rule) applies the same adaptive layer to EBB only where a validation-tail rule inside the initialization window selects it, and is not ranked. Bold marks the column’s best ranked method. Full tables with all quantiles, coverage, significance, and cost are in Appendix E and Appendix F.

Method	Fixed origin					Walk-forward				
	OR	M5	Auto	Carp.	RAF	OR	M5	Auto	Carp.	RAF
Best classical or static-conformal baseline	0.985	1.690	0.319	0.356	0.310	1.059	1.546	0.314	0.349	0.309
ACI-ADIDA	—	—	—	—	—	0.911	1.419	0.315	0.342	0.308
TweedieGP (per-series GP)	0.884	1.623	0.298	0.321	0.276	0.853	—	0.292	0.312	0.280
EBB	0.885	1.671	0.294	0.323	0.268	0.854	1.485	0.291	0.315	0.268
EBB (calibration by rule)	—	—	—	—	—	0.854	1.379	0.290	0.315	0.268

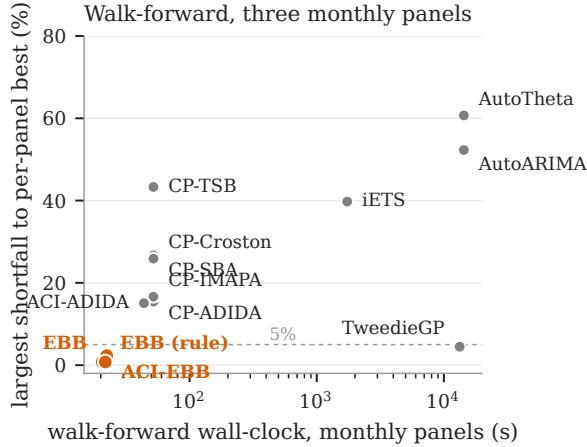


Figure 4: **Largest shortfall against cost under walk-forward updating.** Each point is one method on the three monthly panels, on which every method has a measured wall-clock: y is its largest mean-SPL shortfall to the per-panel best, x the total walk-forward wall-clock (TweedieGP uses twelve workers, every other method one process). The fixed-origin counterpart and the full regret tables are in Appendix E.

error criteria should not be the sole basis for comparing intermittent-demand forecasters, and we use them only as diagnostics.

6 Limitations

(1) **The remaining gap to a per-series specialist.** The log-normal size block keeps item posteriors analytic, and official TweedieGP is better on Online Retail, Carparts, and M5 at fixed origin and on On-

line Retail and Carparts under walk-forward updating, by under one percent except on M5 (2.9%). Three model-side remedies were tested and none closes it (Appendix F.4): heavier-tailed size laws move mean SPL by at most 0.003, item-specific forgetting rates improve M5 by 1.2% but leave it 1.8% behind, and a separate occurrence discount moves M5 by 0.14%. The gap lies in the posterior, where the specialist adapts each series in time with one latent process for occurrence and size, not in the predictive law’s shape or the forgetting schedule.

(2) **What the resolution condition is, and what it is not.** Corollary 1 uses a Gaussian working model and a χ^2 approximation, accurate except for small groups with strongly unequal counts, where it understates the collapse probability by up to 0.21; it is an estimator-specific estimability statement, not an information-theoretic bound. Its calibration on the controlled surface is simulation-based, and the room it assigns to a real panel is read within the panel’s forgetting regime with an interpolation across two series lengths. The pre-fit screen interpolates across regimes it has seen and does not extrapolate across the discount, and its real-series transfer test could check only rejections, since an injected partition never earns material room once a single pool has absorbed the heterogeneity. On the one panel with room, Carparts, neither the real nor the simulated learned partition collects it, and neither label sample-splitting nor credibility shrinkage of the group variance recovers the loss in simulation (Appendix F.8); a partition objective that maximizes resolvable separation rather than marginal likelihood is the implicated next step.

(3) **Scope.** We do not run the external hierarchical Bayesian implementations we cite; a same-model

Gibbs sampler matches plug-in EB within 0.1% (Appendix F.15), so the inference half of that gap is closed and the model-class half is not. The adaptive conformal comparison is one ACI design; given how much it moves the baseline, the remaining designs deserve more attention. Selection uses one pinball score; the discount selected at fixed origin is reused under walk-forward updating without reselection; DeepState, Chronos-Bolt, and Deep Renewal Process comparisons use subset or proxy protocols (Appendix F.13); CRPS and a cost-based criterion remain unrun.

7 Conclusion

Forgetting and pooling are one decision. In a hierarchical empirical-Bayes hurdle model with a single recency operator, the credibility weight an item retains after discounting governs both how much a shared prior can move its forecast and how well the data can resolve the structure of that prior; a two-sided bound makes the trade-off explicit and its corollary makes it computable before a partition is fitted. On five public panels the diagnostic is right in both directions. Where credibility is low the shared prior pays, and it pays more the shorter the histories become. Where it is high, refining the pooling structure is worth nothing and nothing is found; on the one panel with room, the mixture objective fails to collect it in data and in simulation alike, which locates the open problem in the objective rather than in the bound. The forecaster that carries the diagnostic selects structure, forgetting, and calibration on one validation tail and is, with a per-series Gaussian-process specialist, the only method within five percent of the best competitor on every panel under both protocols, at a small fraction of the cost. Two conventions of the field obscure results of this kind: absolute-error metrics that reward the all-zero forecast, and static conformal wrappers that understate what conformal prediction can do. The next steps follow from the gap the diagnostic exposes: partition objectives that optimize resolvable separation, and decision-aware sequential selection.

References

- [1] H. Bühlmann. Experience rating and credibility. *ASTIN Bulletin*, 4(3):199–207, 1967.
- [2] N. Chapados. Effective Bayesian modeling of groups of related count time series. In *ICML*, 2014.
- [3] D. Chen. Online Retail dataset, UCI Machine Learning Repository, 2015.
- [4] A. Cini, D. Mandic, and C. Alippi. Graph-based time series clustering for end-to-end hierarchical forecasting. In *ICML*, volume 235 of PMLR, pages 8985–8999, 2024.
- [5] J. D. Croston. Forecasting and stock control for intermittent demands. *Operational Research Quarterly*, 23(3):289–303, 1972.
- [6] S. Damato, D. Azzimonti, and G. Corani. Forecasting intermittent time series with Gaussian processes and Tweedie likelihood. *International Journal of Forecasting*, 2025.
- [7] I. Gibbs and E. Candès. Adaptive conformal inference under distribution shift. In *NeurIPS*, 2021.
- [8] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *JASA*, 102(477):359–378, 2007.
- [9] J. G. Ibrahim and M.-H. Chen. Power prior distributions for regression models. *Statistical Science*, 15(1):46–60, 2000.
- [10] C. Keribin. Consistent estimation of the order of mixture models. *Sankhyā A*, 62(1):49–66, 2000.
- [11] S. Kolassa. Evaluating predictive count data distributions in retail sales forecasting. *International Journal of Forecasting*, 32(3):788–803, 2016.
- [12] N. Kourentzes and F. Petropoulos. Forecasting with multivariate temporal aggregation. *Int. J. Production Economics*, 181:145–153, 2016.
- [13] J. Lei, M. G’Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman. Distribution-free predictive inference for regression. *JASA*, 113(523):1094–1111, 2018.
- [14] X. Liu, J. Liu, G. Woo, T. Aksu, Y. Liang, R. Zimmermann, C. Liu, J. Li, S. Savarese, C. Xiong, and D. Sahoo. Moirai-MoE: Empowering time series foundation models with sparse mixture of experts. In *ICML*, volume 267 of PMLR, pages 38940–38962, 2025.
- [15] S. Makridakis, E. Spiliotis, and V. Assimakopoulos. M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting*, 38(4):1346–1364, 2022.
- [16] P. Montero-Manso and R. J. Hyndman. Principles and algorithms for forecasting groups of time series: Locality and globality. *International Journal of Forecasting*, 37(4):1632–1653, 2021.
- [17] K. Nikolopoulos, A. A. Syntetos, J. E. Boylan, F. Petropoulos, and V. Assimakopoulos. An aggregate-disaggregate intermittent demand approach (ADIDA). *JORS*, 62(3):544–554, 2011.
- [18] J. Pitkin, I. Manolopoulou, and G. Ross. Bayesian hierarchical modelling of sparse count processes in retail analytics. *Annals of Applied Statistics*, 18(2):946–965, 2024.

- [19] D. Salinas, V. Flunkert, J. Gasthaus, and T. Januschowski. DeepAR: Probabilistic forecasting with autoregressive recurrent networks. *International Journal of Forecasting*, 36(3):1181–1191, 2020.
- [20] M. Seeger, D. Salinas, and V. Flunkert. Bayesian intermittent demand forecasting for large inventories. In *NeurIPS*, 2016.
- [21] R. D. Snyder, J. K. Ord, and A. Beaumont. Forecasting the intermittent demand for slow-moving inventories: A modelling approach. *International Journal of Forecasting*, 28(2):485–496, 2012.
- [22] I. Svetunkov and J. E. Boylan. iETS: State space model for intermittent demand forecasting. *Int. J. Production Economics*, 265, 2023.
- [23] A. A. Syntetos and J. E. Boylan. The accuracy of intermittent demand estimates. *International Journal of Forecasting*, 21(2):303–314, 2005.
- [24] A. A. Syntetos, J. E. Boylan, and J. D. Croston. On the categorization of demand patterns. *JORS*, 56(5):495–503, 2005.
- [25] R. H. Teunter, A. A. Syntetos, and M. Z. Babai. Intermittent demand: Linking forecasting to inventory obsolescence. *EJOR*, 214(3):606–615, 2011.
- [26] M. West and J. Harrison. *Bayesian Forecasting and Dynamic Models*. Springer, 2nd edition, 1997.
- [27] A. C. Türkmen, Y. Wang, and T. Januschowski. Intermittent demand forecasting with deep renewal processes. *arXiv:1911.10416*, 2019.
- [28] T. Januschowski, J. Gasthaus, Y. Wang, D. Salinas, V. Flunkert, M. Bohlke-Schneider, and L. Callot. Criteria for classifying forecasting methods. *International Journal of Forecasting*, 36(1):167–177, 2020.
- [29] H. Hewamalage, C. Bergmeir, and K. Bandara. Global models for time series forecasting: A simulation study. *Pattern Recognition*, 124:108441, 2022.

A Notation

$y_{i,t}, z_{i,t}, \ell_{i,t}$	demand, occurrence indicator, log-size
T_i	initialization-window length of item i
$g(i), K$	pooling group of item i ; number of mixture components
α_g, β_g	Beta occurrence prior; $\phi_g = \alpha_g + \beta_g$ its strength
$\mu_{0,g}, \tau_g^2, \sigma_g^2$	size prior location, between/within-item variances
$w; b$	recency discount; forgetting strength $b = 1 - w$
$n_i(w), c_i(w)$	discounted observation / occurrence pseudo-counts
$\bar{p}_i(w)$	discounted empirical occurrence rate $c_i(w)/n_i(w)$
$\lambda_i(w)$	credibility weight $n_i(w)/(n_i(w) + \phi_g)$
$\hat{\pi}_i, \hat{\mu}_i, \hat{\sigma}_i^2$	posterior means of occurrence, log-size, variance
α_p	classical TSB occurrence smoothing constant ($= 1 - w$)
δ	occurrence drift rate (Proposition 3)
M, n	candidate count and validation size (Prop. 4)
B	bootstrap draws for hyperparameter averaging

B Supporting Results

This appendix records four connections that support the construction but are not the paper’s main theoretical contribution: the relation to TSB, a tracking bound under drift, finite-class selection, and classical credibility.

B.1 Relation to the TSB occurrence recursion

Write $p_t = z_{i,t}$ for one fixed item. Let $\hat{p}_T$ be classical TSB’s occurrence estimate with smoothing $\alpha_p = 1 - w$ and initialization $\hat{p}_0$.

Proposition 2 (TSB occurrence as a boundary case). For $w \in (0, 1)$: (i) $\lim_{(\alpha, \beta) \rightarrow (0, 0)} \hat{\pi}(w) = \bar{p}(w)$; (ii) $\hat{p}_T = (1 - w) \sum_{t=1}^T w^{T-t} p_t + w^T \hat{p}_0$ and $\bar{p}(w) = (\hat{p}_T - w^T \hat{p}_0)/(1 - w^T)$; (iii) hence $\bar{p}(w) = \hat{p}_T$ under the self-consistent initialization $\hat{p}_0 = \bar{p}(w)$, and $|\bar{p}(w) - \hat{p}_T| \leq w^T$ in general.

The equivalence concerns only the occurrence block. Classical TSB uses arithmetic smoothing for positive sizes, whereas EBB discounts positive log-sizes under (1).

B.2 Tracking bound under occurrence drift

Proposition 3 (Tracking risk under drift). If $p_t \sim \text{Bern}(\pi_t)$ independently and $|\pi_t - \pi_T| \leq \delta(T - t)$, then $|\mathbb{E}\bar{p}(w) - \pi_T| \leq \delta w / [(1 - w)(1 - w^T)]$ and $\text{Var}\bar{p}(w) \leq \frac{1}{4}(1 - w)(1 + w^T) / [(1 + w)(1 - w^T)]$, with the corresponding bias-variance decomposition for the posterior mean.

Corollary 2. Balancing the two bounds gives $1 - w^* \asymp \delta^{2/3}$, worst-case MSE $O(\delta^{2/3})$, and $w^* \rightarrow 1$ as $\delta \rightarrow 0$.

This result motivates including $w = 1$ in the candidate grid; it does not claim that chronological validation estimates the bound-optimal w^* .

B.3 Finite candidate selection

Proposition 4 (Conditional finite-class bound). Let $R(c)$ be the expected validation risk of candidate c and $\hat{R}(c)$ its average over n validation items. If the item losses are independent conditional on the fitted candidates and lie in $[0, B]$, then the minimizer $\hat{c}$ over M candidates satisfies, with probability at least $1 - \eta$,

$$R(\hat{c}) \leq \min_c R(c) + B \sqrt{\frac{2 \log(2M/\eta)}{n}}.$$

The independent units are items, because observations and quantiles are averaged within item before selection. The released selector does not clip scaled pinball loss, so the bounded-loss assumption is not automatic and this proposition is not presented as a formal guarantee for the unclipped implementation. It records the standard finite-class behavior under an explicit clipping condition.

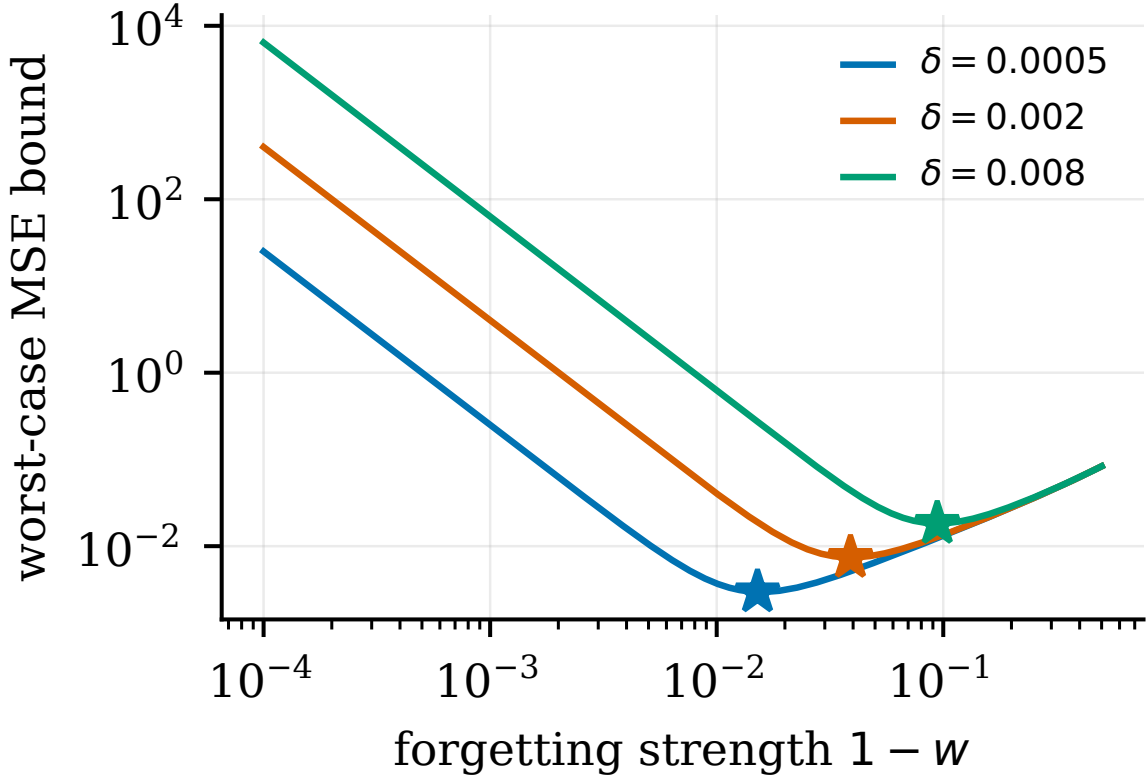


Figure 5: Bias–variance bounds in Proposition 3. Stars mark the minimizing forgetting strength; as drift δ increases, the optimum moves at the rate in Corollary 2.

B.4 Classical credibility

Proposition 5 (Pooling reduces Bayes risk in the stationary model). Under the undiscounted Beta–Binomial occurrence model with known hyperparameters, the posterior mean is the Bayes estimator of π_i under squared loss and has strictly smaller Bayes risk than the per-item empirical rate whenever $\phi_g > 0$. Its optimal weight is the classical credibility factor $\lambda_i = n_i / (n_i + \phi_g)$ [1].

This is the Bühlmann–Straub result specialized to Bernoulli occurrences. It justifies pooling when the stationary hierarchical model is correctly specified; it does not assert that a misspecified or stale group prior helps under drift.

C Proofs

C.1 Proof of Proposition 2

(i) Immediate from (6) since $c(w), n(w)$ are finite. (ii) Induction: $\hat{p}_1 = (1 - w)p_1 + w\hat{p}_0$. If the closed form holds at $T - 1$, then $\hat{p}_T = (1 - w)p_T + w\hat{p}_{T-1} = (1 - w)\sum_{t=1}^T w^{T-t}p_t + w^T\hat{p}_0$. With $n(w) = (1 - w^T)/(1 - w)$,

$$\bar{p}(w) = \frac{(1 - w)\sum_{t=1}^T w^{T-t}p_t}{1 - w^T} = \frac{\hat{p}_T - w^T\hat{p}_0}{1 - w^T}.$$

(iii) Rearranging, $\bar{p}(w) - \hat{p}_T = w^T(\bar{p}(w) - \hat{p}_0)$, and $|\bar{p}(w) - \hat{p}_0| \leq 1$. □

C.2 Proof of Proposition 6

$n(w) = (1 - w^T)/(1 - w) \leq 1/(1 - w)$ and $x \mapsto x/(x + \phi)$ is increasing, so $\lambda(w) \leq [1/(1 - w)]/[1/(1 - w) + \phi] = 1/(1 + (1 - w)\phi)$. $\square$

C.3 Proof of Proposition 3 and Corollary 2

With $k = T - t$ and $\sum_{k \geq 0} kw^k = w/(1 - w)^2$,

$$|\mathbb{E}\bar{p}(w) - \pi_T| = \left| \frac{\sum_t w^{T-t}(\pi_t - \pi_T)}{n(w)} \right| \leq \frac{\delta w/(1 - w)^2}{(1 - w^T)/(1 - w)} = \delta \frac{w}{(1 - w)(1 - w^T)}.$$

By independence and $\pi_t(1 - \pi_t) \leq \frac{1}{4}$,

$$\text{Var } \bar{p}(w) \leq \frac{\frac{1}{4}(1 - w^{2T})/(1 - w^2)}{[(1 - w^T)/(1 - w)]^2} = \frac{1}{4} \cdot \frac{(1 - w)(1 + w^T)}{(1 + w)(1 - w^T)}.$$

Decomposing $\hat{\pi} - \pi_T = \lambda(\bar{p} - \mathbb{E}\bar{p}) + \lambda(\mathbb{E}\bar{p} - \pi_T) + (1 - \lambda)(\mu_\pi - \pi_T)$, the first term is zero-mean and independent of the deterministic remainder, giving $\mathbb{E}(\hat{\pi} - \pi_T)^2 \leq [\lambda B(w) + (1 - \lambda)|\mu_\pi - \pi_T|]^2 + \lambda^2 \text{Var } \bar{p}(w)$ with $B(w)$ the bias bound. For the corollary set $b = 1 - w$ and let $T \rightarrow \infty$ (so $w^T \rightarrow 0$): the squared bias bound behaves as $\delta^2(1 - b)^2/b^2 \sim \delta^2/b^2$ and the variance bound becomes $\frac{1}{4}b/(2 - b)$, which is $b/8$ to leading order as $b \rightarrow 0$. Minimizing $f(b) = \delta^2/b^2 + b/8$ gives $f'(b) = -2\delta^2/b^3 + 1/8 = 0$, i.e., $b^* = (16\delta^2)^{1/3} = 2^{4/3}\delta^{2/3}$, and $f(b^*) = O(\delta^{2/3})$. $\square$

C.4 Proof of Proposition 4

For each candidate c , Hoeffding's inequality for independent scores in $[0, B]$ gives $\Pr(|\hat{R}(c) - R(c)| > t) \leq 2\exp(-2nt^2/B^2)$. A union bound over the M candidates makes $\sup_c |\hat{R}(c) - R(c)| \leq t$ hold with probability at least $1 - 2M\exp(-2nt^2/B^2)$. Setting the right side to η yields $t = B\sqrt{\log(2M/\eta)/(2n)}$. On this event, $R(\hat{c}) \leq \hat{R}(\hat{c}) + t \leq \hat{R}(c^*) + t \leq R(c^*) + 2t$ with $c^* = \arg \min_c R(c)$, which is the claim with constant $2t = B\sqrt{2\log(2M/\eta)/n}$. $\square$

C.5 Proof of Proposition 5

With known (α_g, β_g) and $s_i \mid \pi_i \sim \text{Bin}(n_i, \pi_i)$, $\pi_i \sim \text{Beta}(\alpha_g, \beta_g)$, the posterior mean $\hat{\pi}_i = \lambda_i \bar{p}_i + (1 - \lambda_i)\mu_\pi$ with $\lambda_i = n_i/(n_i + \phi_g)$ minimizes Bayes risk under squared loss among all estimators (tower property). Restricting to linear combinations $a\bar{p}_i + b$, the Bayes risk is $a^2 \mathbb{E}[\text{Var}(\bar{p}_i \mid \pi_i)] + (1 - a)^2 \text{Var}(\pi_i)$ at the optimal intercept, minimized at $a^* = \text{Var}(\pi_i)/(\text{Var}(\pi_i) + \mathbb{E}[\text{Var}(\bar{p}_i \mid \pi_i)]) = \lambda_i$, the Bühlmann credibility weight. The minimum is strictly below the risk of $a = 1$ (the empirical rate) whenever $\text{Var}(\pi_i) < \infty$ and $\phi_g > 0$. $\square$

C.6 Proof of Proposition 1 and Corollary 1

(i) $\lambda_i = n_i/(n_i + \kappa_g)$ is increasing in n_i , and $n_i \leq \sum_{k \geq 0} w^k = (1 - w)^{-1}$ for $w < 1$. Substituting the bound, $\lambda_i \leq \frac{(1-w)^{-1}}{(1-w)^{-1} + \kappa_g} = [1 + (1 - w)\kappa_g]^{-1}$, which is < 1 whenever $w < 1$ and $\kappa_g > 0$.

(ii) Write $\bar{y}_i = \mu_i + e_i$ with $\mu_i \sim N(\mu_g, \tau_g^2)$ and $e_i \sim N(0, s_i)$ independent, $s_i = \sigma_g^2/n_i$, so $\text{Var}(\bar{y}_i) = \tau_g^2 + s_i$. For the sample variance $S_g^2 = (m_g - 1)^{-1} \sum_{i \in g} (\bar{y}_i - \bar{\bar{y}})^2$, independence gives $\text{Var}(\bar{\bar{y}}) = m_g^{-2} \sum_i (\tau_g^2 + s_i)$ and hence

$$\mathbb{E}[S_g^2] = \frac{\sum_i (\tau_g^2 + s_i) - m_g \text{Var}(\bar{\bar{y}})}{m_g - 1} = \frac{(1 - m_g^{-1}) \sum_i (\tau_g^2 + s_i)}{m_g - 1} = \tau_g^2 + \bar{s}_g,$$

exactly, with no balance assumption. The estimator truncates precisely when $S_g^2 \leq \bar{s}_g$. Under the working model S_g^2 is approximately $(\tau_g^2 + \bar{s}_g)\chi_{m_g-1}^2/(m_g - 1)$, so $\text{Var}(S_g^2) \simeq \frac{2}{m_g-1}(\tau_g^2 + \bar{s}_g)^2$ and the standardized distance from the truncation boundary is

$$\frac{\mathbb{E}[S_g^2] - \bar{s}_g}{\sqrt{\text{Var}(S_g^2)}} = \frac{\tau_g^2}{\sqrt{2/(m_g - 1)}(\tau_g^2 + \bar{s}_g)} = \sqrt{\frac{m_g - 1}{2}} \rho_g,$$

giving the stated normal approximation for $\Pr[S_g^2 \leq \bar{s}_g]$. The discount bound follows from $n_i \leq (1-w)^{-1}$: $s_i = \sigma_g^2/n_i \geq (1-w)\sigma_g^2$ for every i , hence $\bar{s}_g \geq (1-w)\sigma_g^2$, and ρ_g is decreasing in $\bar{s}_g$. Finally $\hat{\tau}_g^2 = 0$ gives $\kappa_g = \infty$ and $\lambda_i = 0$.

Corollary. Requiring the collapse probability to stay below a fixed level is, by (ii), the condition $\rho_g \sqrt{(m_g - 1)/2} \geq c$ for a constant c of order one. Solving $\rho_g = \tau_g^2/(\tau_g^2 + \bar{s}_g) \geq c\sqrt{2/(m_g - 1)}$ for τ_g^2 gives $\tau_g^2 \geq \bar{s}_g c\sqrt{2}/(\sqrt{m_g - 1} - c\sqrt{2})$, i.e. the stated order $\bar{s}_g/(\sqrt{m_g - 1} - 1)$ up to the constant. Refining a partition decreases m_g and, by construction of the mixture objective, τ_g^2 , while decreasing w raises the floor $(1-w)\sigma_g^2$ on $\bar{s}_g$. Each shrinks the left side or raises the right side. $\square$

The algebraic steps, the exact unbiasedness in (ii), and the monotonicity claims are verified with a computer algebra system in the released test suite. The χ^2 approximation in (ii) is checked against Monte Carlo simulation, agreeing to within 0.05 in probability across the regimes we use.

D Implementation details

EB fitting. Group Beta–Binomial hyperparameters maximize the (weighted) marginal likelihood by L-BFGS-B with positivity bounds; the size block uses pooled within-item variance for σ_g^2 and an intercept-only restricted marginal likelihood in τ_g^2 with profiled $\mu_{0,g}$; item-level process variances use the conjugate scaled-inv- χ^2 update with $\nu=20$. With discounting, every count is replaced by its weighted analogue (4); the Beta–Binomial marginal extends to real pseudo-counts via log-Gamma functions. The predictive law is zero-inflated log-normal with predictive variance $\hat{\sigma}_{\text{proc}}^2 + \hat{v}_\mu$. The released configuration averages quantiles over $B=20$ bootstrap resamples of the item-statistics rows. An optional location-scale calibration layer exists but is *off* in all headline results; the adaptive conformal comparison is reported as a separate baseline rather than used to modify EBB after fitting.

Mixture EM. Deterministic quantile initialization over (occurrence rate, mean log-size); degenerate components (mass < 2) frozen to global fallback priors; tolerance 10^{-5} , max 60 iterations; $K \in \{1, \dots, 6\}$ by BIC with $5K + (K - 1)$ parameters.

Selection. Internal chronological split (first 80% / last 20%, min head 8); score = mean pinball over $q \in \{0.5, 0.75, 0.9\}$, per-series scaled by head mean absolute demand, averaged within item and then uniformly across items; grid over $w \in \{1, 0.999, 0.997, 0.995, 0.99, 0.98, 0.95, 0.90\}$ and partitions {global, taxonomy, mixture(BIC)}. Selected configurations per dataset are reported beneath Table 10; diagnostics (`remix_selection_diagnostics.csv`) accompany every run.

Environment and timing methodology. All experiments ran on one Windows 11 workstation (Intel Core Ultra 9 275HX, 32 GB RAM). Core determinism tests are bit-identical within this environment; cross-environment bit identity is not claimed. Timing is wall-clock and excludes data loading. Joint selection over the 24 candidates takes 51–243 seconds per panel, with one deterministic six- K EM/BIC search for the mixture on the fitting head. Final model-only point plus five-quantile prediction with $B = 20$ takes 6.2 seconds on Carparts, 38.0 on Online Retail, and 124.0 on M5, whose external span has 3.235 million observations. GPU is not used in EBB.

Baselines. StatsForecast defaults; TSB $(\alpha_d, \alpha_p) = (0.5, 0.45)$; AutoARIMA/AutoTheta season 7 (daily) / 12 (monthly) for intervals. TSB-tuned: 5×5 grid $\alpha_d \in \{0.05, 0.1, 0.2, 0.35, 0.5\}$, $\alpha_p \in \{0.01, 0.05, 0.1, 0.25, 0.45\}$ (25 candidates, the same order as EBB’s 24), scored on the identical 80/20 chronological split by per-series scaled MAE (so neither method is scored on its home criterion), refit on the full window. Selected (α_d, α_p) : Online Retail (0.10, 0.25); M5 (0.20, 0.01); Auto (0.20, 0.10); Carparts (0.50, 0.45), i.e. the textbook constants; RAF (0.50, 0.25). Static conformal wrappers use a chronological 80/20 split and a *panel-pooled* signed-residual distribution; quantiles are zero-truncated and monotonized. The ACI variants retain that residual pool and adapt the requested level for every series–quantile pair with $\gamma = 0.05$. Tweedie-GLM is a per-series autoregressive GLM with log link, $p=1.5$, 14 lags (daily) / 6 (monthly), and $\ell_2=0.1$. TweedieGP uses the authors’ official implementation with Tweedie likelihood, median-demand scaling, RBF kernel, and 50,000 predictive samples; zero-only training series receive a zero fallback so that every series is scored. iETS uses `smooth::adam` with automatic occurrence modeling.

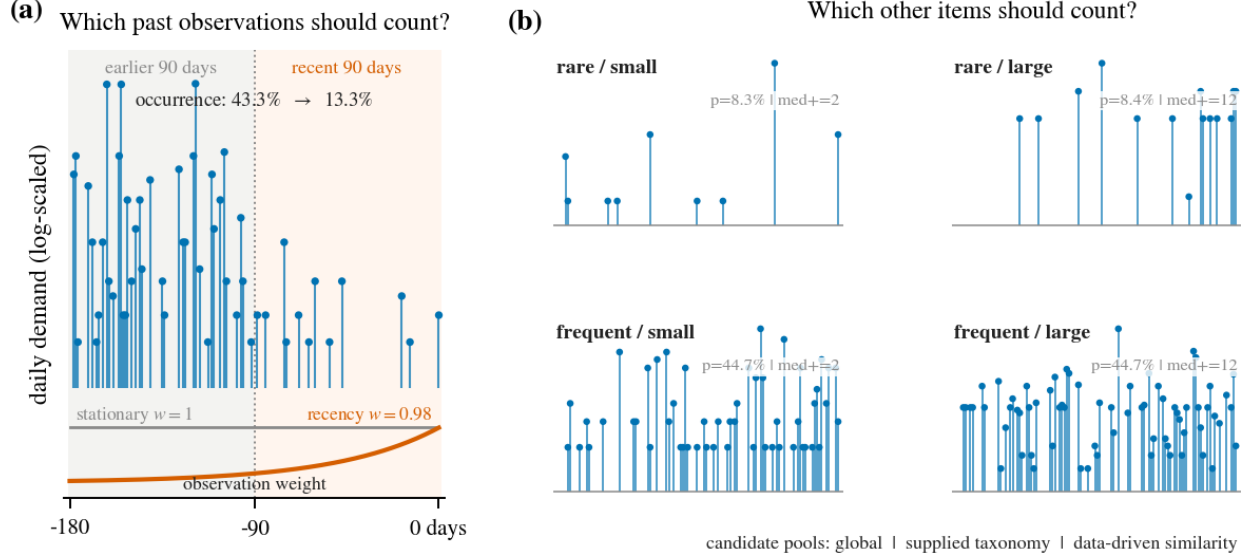


Figure 6: The two coupled choices, in real intermittent demand. **(a)** One item’s daily history: occurrence falls from 43.3% to 13.3% between the earlier and the recent ninety days, so a stationary weighting ($w = 1$) and a recency weighting ($w = 0.98$) summarize the same history differently—which past observations should count? **(b)** Four items from the same panel, spanning rare and frequent occurrence at small and large sizes: any pooled prior must decide which of these heterogeneous neighbors an item should borrow strength from—which other items should count? Candidate pools are a global pool, a supplied taxonomy, and data-driven similarity.

E Main-text material in full

This appendix keeps, verbatim, the tables, figures, and paragraphs of the long-form manuscript that the main text summarizes.

E.1 The two coupled choices in real data

E.2 Persistent prior leverage

Fix an item and suppress its subscripts. Because $n(w) = \sum_{t=1}^T w^{T-t} \leq (1-w)^{-1}$ for $w < 1$, the occurrence credibility weight cannot converge to one.

Proposition 6 (Forgetting keeps pooling relevant). For every horizon T , $\lambda(w) \leq \{1 + (1-w)\phi\}^{-1} < 1$ when $w < 1$, whereas $\lambda(1) \rightarrow 1$ as $T \rightarrow \infty$. Hence the pooled occurrence prior retains positive weight indefinitely under forgetting.

The statement is about leverage, not benefit: it does not guarantee that a particular prior is a good shrinkage target. It says that the choice of target can continue to affect forecasts even for old series.

E.3 Resolution diagnostics recorded by the selector

Fitting each candidate already produces everything the resolution analysis of Section 4.2 needs, so the selector records it rather than recomputing it later. For every (r, w) and every resulting group g we store, from the fitting head only, the median occurrence and positive-size credibility weights $\lambda_i^{(o)}$ and $\lambda_i^{(+)}$, the share of items with $\lambda_i^{(+)} < 0.1$, the group sizes m_g , and the resolution statistic

$$\hat{z}_g = \frac{S_g^2 - \bar{s}_g}{\sqrt{2/(m_g - 1) S_g^2}},$$

Table 2: Pooling candidates used by EBB, ordered coarse to fine. Every data-dependent label is computed from the selector’s fitting head, never its validation tail.

Candidate	Shared-prior groups
Global	All items share one prior.
Taxonomy	Four ADI/CV ² demand-pattern classes (smooth, intermittent, erratic, lumpy) using the classical thresholds 1.32 and 0.49 [24].
Learned	Hard assignments from a mixture of occurrence–size prior components; $K \in \{1, \dots, 6\}$ is selected by BIC.

whose nonpositive values coincide with the collapse event $\hat{\tau}_g^2 = 0$ of Proposition 1. These quantities cost nothing beyond the fit that produced them, they are computed before any external observation is seen, and they answer a question the validation score cannot: not which candidate scores best on the tail, but whether the discounted data contain enough information to estimate the candidate’s group structure at all. Section E.5 uses them to separate the two. The full selection surface and the accompanying diagnostics are released with every run (`remix_selection_diagnostics.csv`).

E.4 Pooling candidates and computational profile

Let $\mathcal{R}$ contain three ways to construct $g(\cdot)$ from the fitting data. Table 2 orders them from coarse to fine: one prior for the panel, one prior per fixed demand-pattern class, and one prior per learned component. The ladder is deliberate. The question this paper asks is not which rung is best in general—that depends on the panel—but whether the discounted data support moving up it at all. Section 4.2 gives the condition under which they do, and Section E.5 reports what happens when they do not.

For the learned candidate, component k carries $\eta_k = (\alpha_k, \beta_k, \mu_{0,k}, \tau_k^2, \sigma_k^2)$. Its item marginal is the product of Beta–Binomial occurrence evidence and Normal random-effects size evidence. EM has an analytic responsibility update; its M-step maximizes responsibility-weighted EB objectives. A deterministic quantile initialization makes repeated fits seed-free. Mixture labels are built once from the fitting head and shared across the discount grid, with $K \in \{1, \dots, 6\}$ selected by BIC at that fit. Re-learning the partition separately at every discount is a natural variant—changing w changes both the item evidence and the components it can support—but we do not evaluate it here. The selected responsibilities are converted to hard labels so all three candidates share the same downstream forecaster.

Two properties of this cost profile matter for the comparisons that follow. First, the whole pipeline is CPU-only and closed form apart from the mixture search: fitting and predicting a panel costs on the order of milliseconds per series, so the full 24-candidate joint selection completes in minutes on a laptop-class machine and no GPU, tuning budget, or convergence diagnostic is required. Second, estimation is exact rather than approximate at every stage, so repeated end-to-end runs are bit-identical in the released test suite. Neither property is an accuracy claim; both are the terms on which the accuracy results in Section 5 should be read, since several of the strongest comparators there cost two to three orders of magnitude more per series and none is reproducible to the bit. Measured wall-clock figures for every method are in Appendix D.

E.5 Fixed-origin results in full

Table 3 sets the empirical position, and it is a narrow one. EBB attains the lowest mean SPL on Auto and RAF, by 1.6% and 3.0% over the official TweedieGP implementation, both significant (Table 23); TweedieGP attains it on Online Retail, Carparts, and M5, by 0.05%, 0.5%, and 2.9%, of which only the M5 margin is significant. Against every other comparator EBB is best on all five panels, by 10.1% over iETS on Online Retail and 1.2% over CP-Croston on M5. Across three independent 5,000-series M5 samples the selected configuration is identical—(mixture, $w=0.95$, $K=6$)—and EBB has mean SPL 1.7501 ± 0.084 against 1.7547 ± 0.065 for CP-Croston, its closest comparator: ahead on two samples, behind by 1.0% on the third, and ahead on average by 0.3%, well within one standard deviation. We report the spread rather than an aggregate win because it is the quantity Section E.8 explains. The explanation is not that these panels leave no room for a hierarchical construction to work; on three of them the credibility weights say otherwise. It is that the room which exists is not converted by a learned partition, which is why the achievable margin is set by the shared predictive law rather than by the

Table 3: **Probabilistic forecasting is the primary comparison.** Scaled pinball loss (SPL; lower is better) is reported as the mean over $q \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$ and at q90. Bold and underline mark the best and second-best non-degenerate methods. M5 reports seed 42 for direct comparability; three-seed robustness is summarized in the text. EBB uses its raw predictive law, with no post-hoc calibration. TweedieGP is the official implementation of Damato et al. at its released defaults on all five panels; Appendix F.7 reports measured cost. The all-zero row is a metric diagnostic and is excluded from ranking.

Model	Online Retail		M5		Auto		Carparts		RAF	
	Mean	q90	Mean	q90	Mean	q90	Mean	q90	Mean	q90
AutoARIMA	1.0069	1.6297	1.7919	2.4148	0.3189	0.2659	0.4084	0.4897	0.4020	0.5910
AutoTheta	1.0055	1.6056	1.7954	2.3123	0.3403	0.2760	0.3925	<u>0.4697</u>	0.4104	0.5956
CP-Croston	1.5502	2.7686	1.6902	2.1551	0.3260	0.3105	0.4012	0.5433	0.3322	0.4927
CP-SBA	1.5492	2.8601	1.6939	2.1736	0.3256	0.3153	0.3982	0.5418	0.3290	0.4920
CP-TSB	1.1744	2.6746	1.8794	2.3148	0.3459	0.3150	0.3783	0.4842	0.3814	0.4986
CP-ADIDA	1.1239	2.5409	1.7150	2.1949	0.3275	0.3082	0.3571	0.4982	0.3097	0.4905
CP-IMAPA	1.1280	2.5506	1.7190	2.1904	0.3279	0.3084	0.3561	0.4908	0.3116	0.4907
Tweedie-GLM	1.1477	1.7458	4.0693	5.2620	0.3221	0.2783	0.4776	0.6897	0.3121	0.5312
iETS	0.9846	1.7544	2.0282	2.9772	0.3448	0.3252	0.4561	0.5646	0.3494	<u>0.4900</u>
TweedieGP	0.8845	1.5218	1.6228	<u>2.1723</u>	<u>0.2985</u>	0.2598	0.3214	0.4600	<u>0.2763</u>	0.5219
Zero (diagnostic)	0.9161	1.6490	1.9590	3.5262	0.5612	1.0102	0.3736	0.6725	0.2669	0.4805
EBB (ours)	<u>0.8849</u>	<u>1.5268</u>	<u>1.6707</u>	2.4758	0.2937	<u>0.2599</u>	<u>0.3229</u>	0.4817	0.2680	0.4856

pooling structure.

The degenerate row exposes what the central quantiles cannot see. On RAF, the all-zero forecast is marginally better through q90 because the true quantiles of most series are zero. At $q \in \{0.95, 0.975, 0.99\}$ it falls behind every intensity-bearing method, and intensity-bearing models become distinguishable from one another (Appendix F.14). TweedieGP is strongest throughout the Carparts far tail and remains competitive on RAF. The probabilistic claim is therefore not that one distribution wins every quantile, but that EBB is the most consistent aggregate performer across panels and remains informative in the region where a zero forecast ceases to be useful.

Point forecasts tell a deliberately weaker story. EBB is best on six of ten MAE/RMSSE cells and top two on seven, but the zero forecast has the lowest MAE on three panels and classical methods lead several remaining cells. We report the complete point table and horizon decomposition in Appendix F.1: they establish competitiveness and rule out a long-horizon averaging artifact, but do not support uniform point-forecast superiority.

E.6 Walk-forward results in full

Table 4 is the deployment-facing test, and it reproduces the fixed-origin picture for the uncalibrated methods. EBB leads on Auto and RAF, by 0.5% and 4.3% over TweedieGP; TweedieGP leads on Online Retail and Carparts by 0.2% and 0.8%. Per-series paired tests (Table 24) cannot separate the two on Online Retail, Carparts, or Auto, and favor EBB on RAF. Over adaptive conformal inference on ADIDA, EBB gains 6.2% on Online Retail and 8.0% on Carparts, beats every static conformal wrapper on all five panels, and loses M5 by 4.4%.

The M5 loss is a calibration effect, and the same layer is available to EBB. ACI-EBB reads EBB’s predictive quantile at the adaptive level instead of the nominal one. On M5 it lowers mean SPL from 1.4851 to 1.3785, 2.9% below ACI-ADIDA and significant; it also leads Auto (0.2901) and brings Carparts to within 0.2% of TweedieGP. On Online Retail and RAF the same layer hurts, by 6.3% and 2.3%, because the per-series coverage signal is too noisy on those panels to improve an already calibrated forecaster. Whether to apply it is therefore a selection problem, and the initialization window already answers it: on the same 80/20 chronological split used to select the discount, EBB is walked forward over the validation tail with and without the layer, and the layer is applied iff its tail pinball is lower (Appendix 25). The rule applies the layer on M5 and Auto and withholds it elsewhere, matching the ex-post better variant on 4 of five panels (Carparts is the exception, where the two are within 0.5% and statistically undecided), and the tail gains it sees track the external ones (+7.8% versus +7.2% on M5, −7.9% versus −6.3% on Online Retail). The resulting row, EBB (rule), is within 0.8% (Carparts) of the best competing method on every panel under walk-forward updating. Two further facts survive the mixed result: under equal calibration EBB beats the ADIDA base on all five panels (undecided only on Online Retail), and

Table 4: **Strict probabilistic walk-forward evaluation on all five panels.** Online Retail and M5 use seven-day blocks; Auto, Carparts, and RAF use monthly blocks. Mean and q90 are scaled pinball losses; all five quantiles, coverage, and interval width are reported in Appendix F.3. Classical wrappers and ACI refit each block. AutoARIMA, AutoTheta, iETS, and TweedieGP refit every four blocks on Online Retail and every block on the monthly panels; on M5 these four were not run under this protocol on cost grounds (dashes). EBB updates sufficient statistics each block; ACI-EBB applies the ACI-ADIDA calibration layer to EBB’s predictive quantiles, and EBB (rule) applies it only where the validation-tail rule selects it, so it equals one of the two rows above it on each panel and is not ranked. Bold and underline rank non-degenerate methods by SPL; on RAF the all-zero forecast is marginally lower than every method through q90. On M5, the longest daily panel, the adaptive conformal layer decides the ranking: ACI-EBB leads, ACI-ADIDA is second, and uncalibrated EBB is third.

Model	Online Retail		Carparts		Auto		RAF		M5	
	Mean	q90	Mean	q90	Mean	q90	Mean	q90	Mean	q90
AutoARIMA	1.0627	1.5225	0.4195	0.4839	0.3139	0.2632	0.4083	0.5920	—	—
AutoTheta	1.0586	1.4835	0.4056	0.4743	0.3224	0.2619	0.4308	0.6233	—	—
iETS	1.1092	1.9404	0.4362	0.5154	0.3286	0.3009	0.3537	0.4902	—	—
CP-TSB	1.3733	2.8422	0.3836	0.4758	0.3297	0.3009	0.3842	0.4941	1.6616	2.0422
CP-ADIDA	1.1576	2.5143	0.3509	0.4880	0.3175	0.2949	0.3095	<u>0.4894</u>	1.5537	1.9629
CP-IMAPA	1.1773	2.5224	0.3492	0.4789	0.3176	0.2948	0.3127	0.4903	1.5457	1.9490
ACI-ADIDA	0.9108	1.6681	0.3417	0.4767	0.3147	0.2900	0.3085	0.5598	<u>1.4192</u>	<u>1.8064</u>
TweedieGP	0.8526	1.3791	0.3121	0.4326	0.2924	0.2539	0.2801	0.5261	—	—
Zero (diagnostic)	0.9161	1.6490	0.3736	0.6725	0.5612	1.0102	0.2669	0.4805	1.9590	3.5262
EBB	<u>0.8540</u>	<u>1.4002</u>	0.3145	0.4578	<u>0.2910</u>	0.2563	0.2681	0.4864	1.4851	1.8797
ACI-EBB	0.9082	1.5623	<u>0.3128</u>	<u>0.4521</u>	0.2901	<u>0.2560</u>	<u>0.2743</u>	0.5175	1.3785	1.6438
EBB (rule)	0.8540	1.4002	0.3145	0.4578	0.2901	0.2560	0.2681	0.4864	1.3785	1.6438

the adaptive layer’s benefit is a property of the panel, not of the forecaster it wraps.

Table 5 and Figure 4 summarize both protocols by the largest shortfall to the per-panel best. Against competing methods, EBB and TweedieGP are the only ones within five percent everywhere (EBB at most 3.0% at fixed origin and 4.6% under walk-forward updating; TweedieGP 3.1% and 4.5%), and every other method is at least 15% behind on some panel. Counting our own wrapper as a competitor, uncalibrated EBB trails ACI-EBB by 7.7% on M5, and EBB (rule) closes that to 0.8%. The two leaders differ in cost. TweedieGP refits per-series variational GPs at every update, so its walk-forward runs took between 36 minutes (Auto) and 2.4 hours (RAF) with twelve workers and project to about 51 hours on M5 at the Online Retail cadence; EBB updates sufficient statistics in closed form and completes each of the five walk-forward runs in under 40 seconds on one core, and ACI-EBB adds a few seconds (Table 29).

The configuration is selected once at the origin and then updated online; repeated online re-selection is outside the present experiment.

E.7 Controlled test of the resolution mechanism

Real data never reveal the true pooling partition, so on their own they cannot distinguish a partition that is not worth having from one we failed to find. Both appear as a small number. We therefore construct panels where the partition is correct by definition, and use them as a reference against which the learned partitions of Section E.8 are measured: what a good partition is worth at a given level of credibility leverage, rather than what our estimator obtains there.

We simulate from the hierarchical hurdle model of (1) with group centers arranged symmetrically around a fixed grand mean, so that increasing separation does not change the panel scale. Each cell compares a single pool against the known true partition under the same predictive representation, with nine paired replicates, and sweeps the two quantities that Section 4.2 identifies as controlling credibility: the discount w and the series length T . The resulting grid spans the leverage range the real panels occupy, from the near-fully-shrunk regime of Carparts to the item-dominated regime of Auto and RAF.

Table 6 gives the result. The true partition improves NLL and log-size MSE in every cell, and pinball loss

Table 5: **Largest shortfall to the per-panel best.** For each method, the maximum over panels of its mean-SPL regret to the best non-degenerate method on that panel (percent; the panel attaining it in parentheses; Carp. is Carparts), and the number of panels on which the method is within 5% of the best. A superscript gives the number of panels evaluated when fewer than five. ACI-EBB and the rule row exist only under walk-forward updating; the rule row is excluded from the per-panel best. Per-panel regrets are in Table 18.

Method	Fixed origin		Walk-forward	
	Max (panel)	$\leq 5\%$	Max (panel)	$\leq 5\%$
AutoARIMA	50.0 (RAF)	0/5	52.3 (RAF) ⁴	0/4
AutoTheta	53.1 (RAF)	0/5	60.7 (RAF) ⁴	0/4
Tweedie-GLM	150.8 (M5)	0/5	—	—
iETS	41.9 (Carp.)	0/5	39.8 (Carp.) ⁴	0/4
CP-Croston	75.3 (OR)	1/5	60.9 (OR)	0/5
CP-SBA	75.1 (OR)	1/5	61.5 (OR)	0/5
CP-TSB	42.3 (RAF)	0/5	61.1 (OR)	0/5
CP-ADIDA	27.1 (OR)	0/5	35.8 (OR)	0/5
CP-IMAPA	27.5 (OR)	0/5	38.1 (OR)	0/5
ACI-ADIDA	—	—	15.1 (RAF)	1/5
TweedieGP	3.1 (RAF)	5/5	4.5 (RAF) ⁴	4/4
EBB	3.0 (M5)	5/5	7.7 (M5)	4/5
ACI-EBB	—	—	6.5 (OR)	4/5
EBB (rule)	—	—	0.8 (Carp.)	5/5

Table 6: Advantage of the *true* partition over a single pool on simulated panels (mean percentage improvement over nine paired replicates; standard error in parentheses). Positive favors the partition. T is series length.

T	w	NLL	log-size MSE	pinball	MAE
30	0.90	2.40 (0.18)	17.15 (2.39)	2.25 (0.61)	-1.72 (0.73)
30	0.95	1.12 (0.39)	9.00 (2.45)	-0.44 (0.58)	-0.17 (0.51)
30	1.00	1.02 (0.08)	8.37 (1.53)	-0.69 (0.32)	-1.95 (0.59)
120	0.90	1.65 (0.22)	18.36 (1.56)	1.57 (0.64)	-3.34 (1.37)
120	0.95	0.96 (0.11)	3.65 (0.77)	0.23 (0.28)	-1.32 (0.73)
120	1.00	0.23 (0.04)	1.13 (0.31)	-0.32 (0.13)	-0.80 (0.11)

whenever forgetting is strong; in a separate length sweep its pinball advantage decays from 1.53% at $T = 12$ to 0.01% at $T = 400$. Both patterns follow the credibility mechanism directly. Discounting or short histories reduce an item’s own weight $\lambda_i^{(+)}$ and open room for a prior to act; long undiscounted histories drive $\lambda_i^{(+)} \rightarrow 1$ and make every partition irrelevant. Stronger forgetting simultaneously makes group variance components harder to estimate, as Proposition 1 predicts, which is why the gains in the $w = 0.90$ rows are bought against a noisier $\hat{\tau}_g^2$ and do not continue to grow without limit. Extending the grid with a separation axis (Appendix F.8; group centers 0 to 3 log-units apart) shows that the return is governed by the forgetting regime as much as by leverage: with no separation in either block the return is within noise of zero; with $w = 0.90$ a true partition earns 1–2.7% from separation 0.5 upward, with $w = 0.95$ it exceeds 1% only at separation 0.5 with long histories ($\lambda^{(+)} \approx 0.4$), and with $w = 1$ it never does. The learned mixture, run on the same simulated panels, is negative at every moderate separation, and the two remedies already in the code base—label sample-splitting and credibility shrinkage of the group variance—do not recover the loss (grid means -0.58%, -0.79%, -0.66%, and -0.56% combined).

MAE moves the other way in all six cells. A known-correct partition, on data generated from the model it describes, is reported as harmful by the metric most commonly used in this literature. We take this as the cleanest available statement of why Section E.5 treats absolute-error criteria as diagnostics rather than as the headline.

What these panels are for. Plotted against median $\lambda^{(+)}$, the cells of Table 6 and the length sweep trace the grey curves of Figure 8: the return to *correct* structure as a function of leverage. Because the partition is supplied rather than estimated, that return is an upper envelope—no procedure that must also learn the grouping can exceed it, and any procedure that falls below it is losing something to estimation rather than to a

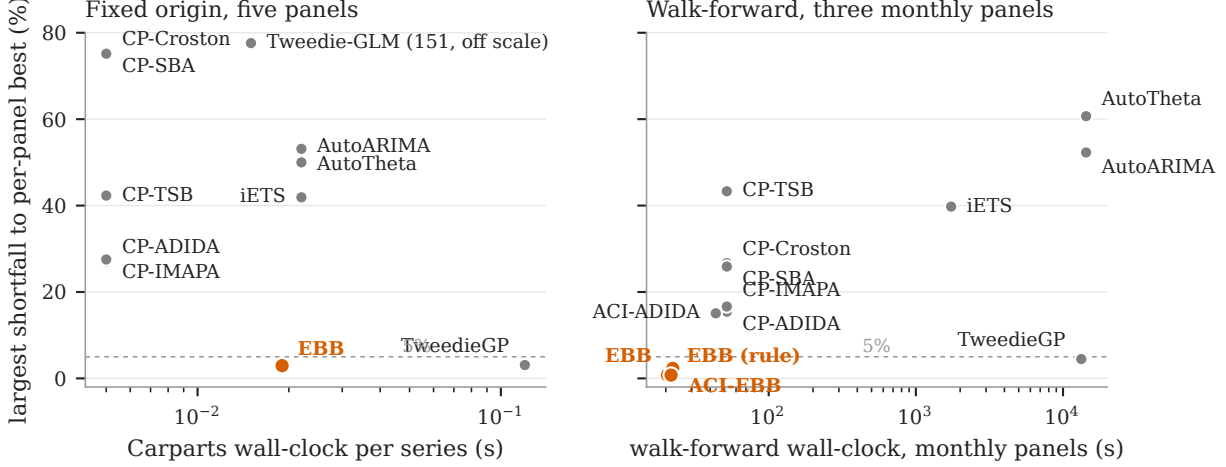


Figure 7: **Largest shortfall against cost.** Each point is one method; y is its largest mean-SPL shortfall to the per-panel best (Table 5), x is measured wall-clock on a log scale. Left: fixed origin over five panels against Carparts cost per series (Table 28). Right: walk-forward over the three monthly panels, on which every method has a measured wall-clock, against the total walk-forward wall-clock on those panels (Table 29; TweedieGP uses twelve workers, every other method one process). The dashed line marks 5%.

lack of structure. This is the comparison Section E.8 draws, and it is the reason the controlled study is necessary rather than decorative: the real panels supply the leverage coordinates, and only a known-partition experiment can supply the ordinate they should be judged against.

Two boundaries apply. The experiment isolates the value of correct structure and says nothing about whether the mixture recovers it—indeed Section E.8 shows it does not, and the gap between the two is precisely what the comparison measures. And it simulates from the model that EBB fits, so the mechanism is tested under correct specification only; a misspecified size law would shift the envelope, though it is hard to see how it would raise it. Full diagnostics, including the growth of $\text{RMSE}(\hat{\tau}_g^2)$ as w falls, are in Appendix F.8.

E.8 Leverage, selection, and ablation

The results so far are consistent and small. This section asks whether their size is a property of the data or of our estimator, and shows that the two can be told apart with quantities the selector already computes on the fitting head (Section 3.4).

What gets selected, and how weakly. All five panels select $w < 1$: Carparts 0.90, Online Retail and M5 0.95, Auto 0.99, and RAF 0.997. Across three independent 5,000-series M5 samples the selected configuration is identical in structure, discount, and mixture order—(mixture, $w=0.95$, $K=6$) in every case.

The pooling axis behaves differently, and the honest summary is that it barely resolves. On three panels the spread between the best and worst pooling structure at the selected discount is under a tenth of a percent of validation SPL: 0.05% on Online Retail, 0.08% on Carparts, 0.02% on RAF. Only two panels separate at all—M5, where the learned mixture beats the global pool by 0.58%, and Auto, where the global pool beats every refinement by 0.31%. Announcing a winner on the other three would report selector noise. What the selector reliably does is choose a discount and avoid poor structural corners; it does not discover demand taxonomies, and we do not claim it does.

Leverage is an upper bound, not a prediction. Table 7 reports the diagnostic. Read as Corollary 1 intends, it does not say which partition wins; it says how much any partition could win by. Two panels have almost no room on the size block: Auto and RAF retain median $\lambda^{(+)} \approx 0.92$ –0.94 even at their selected discounts, with no item below 0.1, so their sub-percent pooling differences are explained before any partition is fitted. The remaining three retain leverage: Carparts is extreme—median 0.224, with 22.5% of items essentially fully

Table 7: **Credibility leverage under a single pool.** Median credibility weight per panel at no forgetting and at the selected discount, for both blocks of the hurdle model: $\lambda_i^{(o)} = n_i/(n_i + \phi_g)$ for occurrence and $\lambda_i^{(+)} = n_i^+/(n_i^+ + \kappa_g)$ for positive size, the latter being the quantity bounded by Proposition 1. A weight near one means the item’s own history dominates its posterior, bounding how far any shared prior can move it. “< 0.1” is the share of items whose size-block weight falls below 0.1. The final column is the external MAE change of a separately selected learned partition relative to the single pool; negative favors the single pool.

Panel	w^*	$\lambda^{(+)} \text{ (size)}$			$\lambda^{(o)} \text{ (occ.)}$		learned vs. pool
		$w=1$	w^*	< 0.1	$w=1$	w^*	
Carparts	0.90	0.802	0.224	22.5%	0.882	0.628	−1.2%
Online Retail	0.95	0.897	0.553	14.6%	0.969	0.895	−2.9%
M5	0.95	0.996	0.841	12.9%	0.998	0.931	−2.3%
Auto	0.99	0.944	0.940	0.0%	0.525	0.422	−3.6%
RAF	0.997	0.926	0.917	0.0%	0.085	0.106	−31.5%

Table 8: Online Retail distributional ablation. Each row is an independent hierarchical run; “auto” denotes its own internal discount selection. The raw predictive law is used and lower is better. Full point metrics are in Appendix Table 11.

Pooling	Forget	SPL	SPL ₉₀	AIW
global	off	0.8899	1.5306	10.51
taxonomy	off	0.8897	1.5299	10.56
learned	off	0.8905	1.5336	10.79
global	auto (.95)	<u>0.8849</u>	<u>1.5268</u>	9.25
taxonomy	auto (.95)	0.8846	1.5254	9.43
learned	auto (.95)	0.8869	1.5343	10.69

shrunk—and Online Retail sits at 0.553 under the selected discount. Leverage alone does not fix the prize, however. The controlled study shows that the true partition’s return at a given $\lambda^{(+)}$ depends on the forgetting regime that produced it and on how far the groups are apart, so the panels are placed on the full surface below.

High leverage is an amplifier with no sign attached. The clearest illustration is RAF, which combines the highest size-block leverage in the panel set ($\lambda^{(+)} = 0.917$) with the lowest occurrence leverage ($\lambda^{(o)} = 0.106$), and loses 31.5% of external MAE to a learned partition. The two facts are one fact. The point forecast is $\hat{y}_i = \hat{\pi}_i \exp(\hat{\mu}_i + \hat{\sigma}_i^2/2)$; when 89% of $\hat{\pi}_i$ comes from the group prior, replacing the group is close to replacing the forecast. This is the empirical content of the caveat attached to Proposition 6: persistent prior leverage guarantees that the choice of target keeps mattering, not that a particular target is good. Read together with the previous paragraph, low leverage caps the upside and high leverage uncaps the downside, and neither makes a learned partition worth using here.

What the ablation confirms. Table 8 is the external counterpart on Online Retail: the three pooling candidates differ by less than 0.1% in mean SPL without forgetting and by less than 0.3% at the selected discount, matching the internal picture. Discounting shows up elsewhere, narrowing the global and taxonomy 80% intervals by 11–12% while improving mean SPL by about half a percent; the full ablation (Appendix Table 11) shows it improves external MAE by 3.5–4.0% and slightly worsens RMSE. The cross-panel result in Table 3 should therefore be attributed to the shared hurdle-EB predictive law and to the temporal half of the selection, not to a learned-partition gain. That is not a concession made after the fact: the leverage diagnostic states the ceiling in advance, and Figure 8 shows we do not reach it.

E.9 Pre-fit screen: protocol, metrics, and real-series transfer

Protocol. The grid crosses separation $\{0, .25, .5, 1, 2\}$, discount $\{1, .95, .90\}$, length $\{30, 60, 120\}$, occurrence mean $\{.1, .25, .5\}$, items per group $\{20, 60\}$, and balanced versus 70/10/10/10 group sizes, with four groups, $\tau^2 = 0.15$, $\sigma^2 = 1$, occurrence separation scaled as $0.8 \min(\text{sep}, 1)$, five replicates per cell, and seed 20260915. Per panel, only the single-pool fit is used for features: median retained leverage on the size and occurrence

Table 9: Held-out screening performance on the simulated grid for the event that the oracle partition’s gain exceeds one percent (base rate 22%). Columns: AUC of the pre-registered R_1 , of R'_1 , and of the logistic screen R_2 on all panels; the same for R'_1 and R_2 on the panels whose single-pool variance component did not collapse; precision and recall of R_2 at its best- F_1 threshold.

Held out	All panels			Resolved			R_2 P / R
	R_1	R'_1	R_2	R'_1	R_2		
hold out a separation level	0.45	0.60	0.84	0.72	0.85	0.54	0.77
hold out a discount	0.64	0.67	0.68	0.66	0.68	0.51	0.49
hold out a length	0.51	0.71	0.82	0.79	0.83	0.54	0.67
random 5-fold	0.51	0.71	0.83	0.78	0.84	0.55	0.65

blocks, $\hat{\tau}^2$, the Beta concentration, the resolution statistic, effective counts, and panel size. The target is the oracle partition’s paired pinball gain over the single pool (estimated hyperparameters, true labels), which can be negative. The pre-registered closed-form score R_1 used an unweighted mean of sampling variances that is dominated by items whose discounted positive count approaches zero; it has no screening power (AUC 0.45–0.64) and is retained in the released artifact. R'_1 replaces that term by the pool’s own weighted estimate $\hat{\tau}^2$; R_2 is a logistic regression on $\log(1 + \cdot)$ of the features, fitted on the training folds only. The design, the amendments, and their order are recorded in `docs/DESIGN_room_diagnostic.md`.

Real-series transfer. Online Retail, Carparts, and the 5,000-series M5 sample keep their zeros, noise, lengths, and scales; a known partition is injected by multiplying every positive value of item i by $\exp(\delta_{g(i)})$ with symmetric offsets $\{\pm 0.5, \pm 1.5\} \times \delta$, $\delta \in \{0, .25, .5, 1, 2\}$, over four randomly assigned groups (three assignments), at the selected discount and again at $w = 0.90$. The oracle gain of the injected partition never exceeds 0.4% on any panel: the single pool absorbs the injected heterogeneity ($\hat{\tau}^2$ rises from 0.04 to 5.1 on Carparts as δ goes from 0 to 2) and the median size-block credibility rises toward one, so the items’ own evidence, not the prior, carries the forecast. R_2 , fitted on the simulated grid and applied unchanged, assigns every injected panel a probability below 0.17 of material room (Figure 10). The transfer test therefore checks the screen’s rejections, not its detections; a real-series test of detections would require panels whose items lack positive observations, which is the short-history regime of Section 5.2.

F Additional results

F.1 Point forecasting and horizon decomposition

Table 10 gives the complete point comparison retained for continuity with the intermittent-demand literature. Rankings exclude the deliberately degenerate Zero and Naive-last rows. EBB takes six of ten MAE/RMSSE cells and is top two on seven, but the result is not uniform: classical methods retain several point-metric wins, and Zero attains the lowest MAE on Online Retail, Carparts, and RAF. This is why the main text uses point accuracy as a secondary diagnostic.

The selected (g, w) pairs are: Online Retail (global, 0.95), M5 (mixture, 0.95), Auto (global, 0.99), Carparts (global, 0.90), and RAF (mixture, 0.997). On three of the five panels the margins between pooling structures on the internal validation surface are under a tenth of a percent, so the structure component of these pairs should be read as Section E.8 reads it: the selector avoids poor structural corners, and no winner is announced where the surface does not resolve one.

F.2 Selector diagnostics

The selection surface is locally flat near its optimum: on Online Retail the best seven candidates span all three pooling structures and lie within 0.51% of one another in validation SPL. Section E.8 reports the credibility leverage and refinement gains that accompany each selected pair, and Figure 8 places them against the controlled panels. The selector is useful for choosing a discount and for avoiding poor structural corners; the winning pooling structure on a panel whose candidates lie within a tenth of a percent should not be read as a scientific taxonomy. Per-panel selection surfaces are released with the code.

Table 10: Point forecasting across the five panels (lower is better; bold best and underline second among non-degenerate methods). Degenerate rows are shown as metric diagnostics and excluded from ranking.

Model	Online Retail		M5		Auto		Carparts		RAF	
	MAE	RMSSE	MAE	RMSSE	MAE	RMSSE	MAE	RMSSE	MAE	RMSSE
CrostonClassic	6.3294	5.1051	1.2345	2.3539	3.4056	1.1488	0.6792	1.0981	2.7315	1.6280
CrostonSBA	6.1953	5.0692	1.2076	2.3517	<u>3.3325</u>	<u>1.1451</u>	0.6628	1.0935	2.6520	1.6272
TSB	<u>5.5736</u>	4.8031	1.2672	2.3612	3.6587	1.2066	0.5396	1.0156	2.1660	1.6570
TSB-tuned	5.6318	4.7985	1.1932	<u>2.3137</u>	3.3917	1.1452	0.5396	1.0156	<u>2.2447</u>	1.6408
ADIDA	5.6860	<u>4.7967</u>	1.2274	2.3398	3.4079	1.1545	0.5556	1.0244	2.2977	<u>1.6245</u>
IMAPA	5.7000	4.7970	1.2317	2.3280	3.4143	1.1555	<u>0.5524</u>	1.0126	2.2850	1.6259
AutoARIMA	6.5494	4.8175	1.2283	2.3169	3.4417	1.1745	0.5822	<u>1.0105</u>	2.3153	1.6262
AutoTheta	8.2862	4.8550	1.2988	2.3143	3.7490	1.2215	0.5704	0.9966	2.3687	1.6399
Zero	4.7549	4.8089	1.2995	2.4629	4.1444	1.4454	0.3867	1.0949	1.1717	1.6345
Naive-last	6.0570	4.8324	1.3750	2.5450	4.3529	1.3818	0.5399	1.1408	1.8340	1.7644
EBB (ours)	5.5325	4.7901	<u>1.2032</u>	2.2938	3.2145	1.1374	0.5726	1.0120	2.3874	1.6245

Table 11: Full Online Retail ablation corresponding to Table 8. Each row is an independent run; “auto” denotes its own internal discount selection. TSB has no native predictive distribution. Lower is better.

Pooling	Forget	MAE	RMSE	RMSSE	SPL	SPL ₉₀	AIW
none (TSB)	fixed	5.5736	18.7272	4.8031	—	—	—
global	off	5.7623	17.6892	4.7851	0.8899	1.5306	10.51
taxonomy	off	5.7663	<u>17.6930</u>	<u>4.7875</u>	0.8897	1.5299	10.56
learned	off	5.8126	17.7064	4.7876	0.8905	1.5336	10.79
global	auto (.95)	5.5325	17.8386	4.7901	<u>0.8849</u>	<u>1.5268</u>	9.25
taxonomy	auto (.95)	<u>5.5624</u>	17.8421	4.7897	0.8846	1.5254	9.43
learned	auto (.95)	5.6925	17.8765	4.7902	0.8869	1.5343	10.69

F.3 Walk-forward point results and protocol

EBB applies the selected discount once to the initialization sufficient statistics and then accumulates revealed blocks without compounding it. On Online Retail, ACI-ADIDA improves the static conformal wrapper substantially and wins q50 by a hair; TweedieGP takes the mean and q90, and EBB takes q75 and is second elsewhere. On Carparts the first two quantiles saturate at zero, EBB leads from q10 through q75, and TweedieGP takes q90 and the mean. On Auto EBB leads q10, q25, and the mean while TweedieGP leads q50 through q90; on RAF EBB leads every quantile, coinciding with the zero forecast through q75; on M5 ACI-EBB leads every quantile from q25 up and the mean, ACI-ADIDA is second, and uncalibrated EBB leads q10. ACI-EBB improves on EBB on M5, Auto, and Carparts and degrades it on Online Retail and RAF (Table 24). Table 18 gives the per-panel regrets behind Table 5. Re-running joint selection at every block is not evaluated.

F.4 Three model-side remedies that do not close the gap

Section E.5 locates the remaining shortfall to TweedieGP on Online Retail, Carparts, and M5. Three remedies within the EBB family are the obvious candidates: a heavier-tailed positive-size law, an item-specific forgetting rate, and a separate forgetting rate for the occurrence block. Both were evaluated on the fixed-origin protocol with the audited pooling structure and, for the size law, the identical fitted posterior; neither changes any panel’s leader.

Predictive-law shape. Table 20 replaces the log-normal predictive law of (3) by log- t , moment-matched Gamma, and inflated-scale alternatives computed from the same item posteriors. Mean SPL moves by at most 0.003 on every panel and in no case reaches TweedieGP.

Item-specific forgetting. Each item receives its own discount w_i , chosen on an inner chronological split of the fitting head by minimizing $n_i^V R_i(w) + \kappa \bar{R}(w)$, where R_i is the item’s validation pinball curve over the grid of Section 3.4, $\bar{R}$ the panel average, and κ an empirical-Bayes shrinkage strength selected on the outer validation tail; $\kappa=\infty$ recovers the panel-wide w^* and $\kappa=0$ leaves each item unshrunk. Table 21 reports the result. The

Table 12: Online Retail point forecasting under strict seven-day walk-forward updates. The hierarchical rows differ in their initialization discount and pooling structure.

Model	MAE	RMSE	RMSSE
CrostonClassic	5.692	16.937	4.854
CrostonSBA	5.586	16.931	4.836
TSB	5.434	17.754	4.815
ADIDA	5.320	16.728	<u>4.637</u>
IMAPA	<u>5.332</u>	<u>16.745</u>	4.632
AutoARIMA	5.482	16.904	5.105
AutoTheta	5.394	16.836	4.735
Stationary hierarchy ($w = 1$)	5.586	17.260	4.725
Forgetting hierarchy (tax., .95)	5.411	17.059	4.682
EBB (selected global, .95)	5.421	17.188	4.696

Table 13: Full Online Retail walk-forward SPL at all five quantiles and on average, with marginal (Cov@80) and positive-demand (Cov⁺@80) coverage and average interval width (AIW). CP and ACI wrappers refit each of the 36 seven-day blocks; AutoARIMA, AutoTheta, iETS, and TweedieGP refit every four blocks; EBB updates sufficient statistics each block. The zero row is excluded from ranking.

Model	q10	q25	q50	q75	q90	Mean	Cov@80	Cov ⁺ @80	AIW
AutoARIMA	0.259	0.572	1.197	1.763	1.522	1.063	0.933	0.762	16.58
AutoTheta	0.233	0.584	1.276	1.717	1.484	1.059	0.931	0.765	16.43
iETS	0.236	0.505	1.302	1.562	1.940	1.109	0.598	0.166	19.74
CP-Croston	0.341	0.778	1.364	1.745	2.632	1.372	0.793	0.478	5.66
CP-SBA	0.334	0.762	1.343	1.725	2.722	1.377	0.793	0.481	5.75
CP-TSB	0.451	0.899	1.289	1.385	2.842	1.373	0.795	0.468	5.71
CP-ADIDA	0.277	0.627	1.050	<u>1.319</u>	2.514	1.158	0.791	0.473	5.47
CP-IMAPA	0.292	0.660	1.083	1.329	2.522	1.177	0.791	0.471	5.47
ACI-ADIDA	0.186	0.468	0.909	1.324	1.668	0.911	0.904	0.646	19.67
ACI-TSB	0.189	0.482	0.948	1.384	1.710	0.943	0.903	0.641	21.21
TweedieGP	<u>0.183</u>	<u>0.458</u>	0.917	1.326	1.379	0.853	0.924	0.709	13.03
Zero (diagnostic)	0.183	0.458	0.916	1.374	1.649	0.916	0.740	0.000	0.00
EBB	0.183	0.458	<u>0.910</u>	1.319	<u>1.400</u>	<u>0.854</u>	0.901	0.621	10.31
ACI-EBB	0.183	0.458	0.915	1.423	1.562	0.908	0.913	0.665	19.55
EBB (rule)	0.183	0.458	0.910	1.319	1.400	0.854	0.901	0.621	10.31

pre-registered adoption criterion (improve Online Retail or M5 by at least 1% without degrading Auto or RAF by more than 0.3%, and either recover Online Retail from TweedieGP or bring M5 within 1.5% of it) fails on its second clause: M5 improves by 1.15% but remains 1.8% behind, and Online Retail is unchanged. Unshrunk selection ($\kappa=0$) is markedly worse on M5, so the shrinkage is necessary; the selected κ is interior on four panels. The per-item option is retained in the released code but is not used in any reported result.

Separate occurrence forgetting. The M5 shortfall sits in the occurrence block (Section E.5), and TSB itself smooths occurrence and size with two constants, so a third remedy gives the occurrence counts their own discount $w^{(o)}$ while the size statistics keep the selected w . $w^{(o)}$ is chosen on the same 80/20 chronological split and criterion as w , from a grid that contains the tied value. Table 22 reports the result. The selector keeps the tied rate on Online Retail, Auto, and RAF, moves M5 by 0.14%, and moves Carparts by 0.77% (to 0.3204, marginally ahead of TweedieGP); the pre-registered criterion of a 1% gain on M5 is not met, so the single rate is retained. A faster occurrence rate ($w^{(o)}=0.8$) removes every zero q90 forecast on M5 and lowers q90 SPL from 2.476 to 2.452 without changing the mean, which locates the M5 tail problem in occurrence persistence rather than in the size law, but the selector does not choose it and we do not override the selector.

F.5 Significance and interval diagnostics

Table 23 tests the headline metric directly: per-series mean SPL, EBB against every comparator of Table 3, with paired t and Wilcoxon signed-rank tests Holm-adjusted within each panel and a comparison counted as decided only when both adjusted tests agree at the 5% level. EBB is significantly better in 43 of the 50 panel-

Table 14: Full Carparts walk-forward SPL at all five quantiles and on average, with marginal (Cov@80) and positive-demand (Cov⁺@80) coverage and average interval width (AIW). Every baseline refits each of the six monthly blocks; EBB updates sufficient statistics. The zero row is excluded from ranking.

Model	q10	q25	q50	q75	q90	Mean	Cov@80	Cov ⁺ @80	AIW
AutoARIMA	0.150	0.282	0.537	0.643	0.484	0.419	0.886	0.620	1.63
AutoTheta	0.120	0.253	0.523	0.657	0.474	0.406	0.902	0.678	1.80
iETS	0.272	0.368	0.493	0.532	0.515	0.436	0.442	0.411	1.17
CP-Croston	0.099	0.245	0.480	0.614	0.539	0.395	0.847	0.580	1.39
CP-SBA	0.098	0.242	0.475	0.611	0.538	0.393	0.847	0.581	1.37
CP-TSB	0.141	0.301	0.487	0.513	0.476	0.384	0.834	0.551	1.27
CP-ADIDA	0.096	0.230	0.411	0.529	0.488	0.351	0.829	0.593	1.28
CP-IMAPA	0.101	0.235	0.413	0.518	0.479	0.349	0.830	0.581	1.27
ACI-ADIDA	0.088	0.221	0.406	0.517	0.477	0.342	0.870	0.604	1.38
TweedieGP	0.075	0.188	0.375	0.490	0.433	0.312	0.931	0.662	1.46
Zero (diagnostic)	0.075	0.187	0.374	0.560	0.672	0.374	0.795	0.000	0.00
EBB	<u>0.075</u>	<u>0.187</u>	<u>0.372</u>	<u>0.481</u>	0.458	0.315	0.932	0.669	1.42
ACI-EBB	0.075	0.187	0.371	0.480	<u>0.452</u>	<u>0.313</u>	0.934	0.676	1.41
EBB (rule)	0.075	0.187	0.372	0.481	0.458	0.315	0.932	0.669	1.42

Table 15: Full Auto walk-forward SPL at all five quantiles and on average, with marginal (Cov@80) and positive-demand (Cov⁺@80) coverage and average interval width (AIW). Every baseline refits each of the six monthly blocks; EBB updates sufficient statistics. The zero row is excluded from ranking.

Model	q10	q25	q50	q75	q90	Mean	Cov@80	Cov ⁺ @80	AIW
AutoARIMA	0.141	0.313	0.447	0.405	0.263	0.314	0.778	0.799	8.73
AutoTheta	0.152	0.320	0.460	0.418	0.262	0.322	0.780	0.825	9.64
iETS	0.172	0.312	0.435	0.423	0.301	0.329	0.564	0.690	7.33
CP-Croston	0.148	0.307	0.437	0.402	0.296	0.318	0.801	0.802	5.97
CP-SBA	0.146	0.304	0.434	0.404	0.301	0.318	0.800	0.803	6.01
CP-TSB	0.155	0.321	0.458	0.414	0.301	0.330	0.808	0.799	5.97
CP-ADIDA	0.149	0.307	0.437	0.400	0.295	0.318	0.802	0.802	5.92
CP-IMAPA	0.149	0.307	0.437	0.400	0.295	0.318	0.802	0.802	5.92
ACI-ADIDA	0.139	0.305	0.438	0.401	0.290	0.315	0.830	0.822	6.74
TweedieGP	0.122	0.279	<u>0.418</u>	0.389	0.254	0.292	0.834	0.828	8.81
Zero (diagnostic)	0.112	0.281	0.561	0.842	1.010	0.561	0.262	0.000	0.00
EBB	0.112	<u>0.275</u>	0.418	0.393	0.256	<u>0.291</u>	0.906	0.873	9.17
ACI-EBB	<u>0.113</u>	0.275	0.416	0.391	<u>0.256</u>	0.290	0.905	0.876	9.41
EBB (rule)	0.113	0.275	0.416	0.391	0.256	0.290	0.905	0.876	9.41

comparator pairs, significantly worse in one (TweedieGP on M5, adjusted $p = 0.003$ and 0.035), and undecided in six. On M5 the mean difference favors EBB against all four conformal wrappers but the paired t does not reach significance after correction (the Wilcoxon strongly rejects symmetry, so the per-series differences are systematic but heavy-tailed); on Carparts neither EBB nor TweedieGP separates from the other (adjusted $p \geq 0.50$ both ways), consistent with the half-percent aggregate margin between them; and on Online Retail TweedieGP is ahead by 0.04% in mean SPL with the paired t at $p = 0.71$. EBB is significantly better than TweedieGP on Auto and RAF.

The paired MAE tests support a systematic improvement over the stationary hierarchy. Against TSB, tuned TSB, ADIDA, and IMAPA, mean MAE favors EBB but the adjusted paired t -tests are not significant; the Wilcoxon tests reverse because those methods win on the typical series while EBB reduces the difficult-series tail. On per-series RMSE, both adjusted tests favor EBB against every comparator at $p < 0.05$. Raw intervals mildly over-cover, most strongly on Carparts.

F.6 Selection objective and efficiency

The selector is not mathematically tied to pinball loss; it scores the criterion supplied on the chronological validation tail. We use item-averaged SPL because the target is a predictive distribution. The best candidates

Table 16: Full RAF walk-forward SPL at all five quantiles and on average, with marginal (Cov@80) and positive-demand (Cov⁺@80) coverage and average interval width (AIW). Every baseline refits each of the twelve monthly blocks; EBB updates sufficient statistics. The zero row is excluded from ranking.

Model	q10	q25	q50	q75	q90	Mean	Cov@80	Cov ⁺ @80	AIW
AutoARIMA	0.066	0.151	0.490	0.742	0.592	0.408	0.944	0.352	8.76
AutoTheta	0.066	0.158	0.497	0.809	0.623	0.431	0.948	0.409	10.03
iETS	0.220	0.270	0.353	0.435	0.490	0.354	0.495	0.019	0.53
CP-Croston	0.090	0.216	0.381	0.471	0.492	0.330	0.828	0.045	1.03
CP-SBA	0.088	0.212	0.375	0.468	0.491	0.327	0.828	0.041	0.98
CP-TSB	0.160	0.317	0.462	0.488	0.494	0.384	0.836	0.025	0.37
CP-ADIDA	0.078	0.187	0.343	0.449	<u>0.489</u>	0.309	0.823	0.021	0.72
CP-IMAPA	0.082	0.192	0.348	0.452	0.490	0.313	0.824	0.019	0.69
ACI-ADIDA	0.057	0.154	0.326	0.446	0.560	0.308	0.909	0.083	1.85
TweedieGP	0.053	0.134	0.270	0.417	0.526	0.280	0.935	0.198	4.07
Zero (diagnostic)	0.053	0.134	0.267	0.400	0.480	0.267	0.919	0.000	0.00
EBB	<u>0.053</u>	<u>0.134</u>	<u>0.267</u>	0.400	0.486	0.268	0.922	0.041	0.95
ACI-EBB	0.053	0.134	0.267	<u>0.401</u>	0.517	<u>0.274</u>	0.927	0.101	2.07
EBB (rule)	0.053	0.134	0.267	0.400	0.486	0.268	0.922	0.041	0.95

Table 17: Full M5 walk-forward SPL at all five quantiles and on average, with marginal (Cov@80) and positive-demand (Cov⁺@80) coverage and average interval width (AIW). Static CP wrappers and ACI-ADIDA refit each of the 93 seven-day blocks on the 5,000-series sample; AutoARIMA, AutoTheta, iETS, and TweedieGP were not run under this protocol on cost grounds; EBB updates sufficient statistics each block. The zero row is excluded from ranking.

Model	q10	q25	q50	q75	q90	Mean	Cov@80	Cov ⁺ @80	AIW
CP-Croston	0.502	1.150	1.917	2.343	1.923	1.567	0.812	0.679	2.26
CP-SBA	0.495	1.137	1.909	2.385	1.956	1.576	0.812	0.684	2.28
CP-TSB	0.531	1.263	2.094	2.377	2.042	1.662	0.831	0.685	2.37
CP-ADIDA	0.479	1.108	1.889	2.330	1.963	1.554	0.827	0.690	2.28
CP-IMAPA	0.480	1.110	1.889	2.301	1.949	1.546	0.828	0.691	2.28
ACI-ADIDA	0.402	<u>0.960</u>	1.718	<u>2.211</u>	<u>1.806</u>	<u>1.419</u>	0.883	0.718	3.48
Zero (diagnostic)	0.392	0.980	1.959	2.938	3.526	1.959	0.606	0.000	0.00
EBB	<u>0.391</u>	0.967	1.852	2.336	1.880	1.485	0.875	0.690	2.77
ACI-EBB	0.389	0.951	<u>1.768</u>	2.140	1.644	1.379	0.891	0.730	3.00
EBB (rule)	0.389	0.951	1.768	2.140	1.644	1.379	0.891	0.730	3.00

are locally close, and a full cross-objective regret matrix remains future work.

Joint selection takes 51–243 seconds across the five panels, excluding data loading, and covers 24 forecast candidates with one deterministic EM/BIC search for the mixture on the fitting head. Final estimation is analytic apart from low-dimensional group optimization; selection and point prediction are deterministic, while probabilistic bootstrap averaging uses the released fixed seed.

F.7 Measured cost of every method

Table 28 reports instrumented wall-clock for every method of Table 3 on Carparts (2,509 series), measured on one CPU workstation in a single session with configurations unchanged. The EBB row includes its complete 24-candidate joint selection in addition to the final fit ($B=20$) and five-quantile prediction; the TSB-tuned row likewise includes its 25-candidate selection. No method uses a GPU. Figure 13 plots cost against mean SPL.

On the daily panels the official TweedieGP implementation was run in full under the authors’ released defaults (sparse variational GP with 200 inducing points and log-spaced initialization for series longer than 200 observations, all training points otherwise; Adam with learning rate 0.1; 50,000 predictive samples; median-demand scaling; restart on numerical failure) and the authors’ daily period. Online Retail (3,649 series) completed with no failed fits at a per-series median of 5.1 s; M5 (5,000 series, 1,294 training observations) completed with no failed fits at a per-series median of 17.3 s, of which roughly half is the 50,000-sample predictive draw over the

Table 18: Per-panel mean-SPL regret (percent) to the best non-degenerate method on that panel, under the fixed-origin and walk-forward protocols. Zero regret marks the panel leader. Dashes mark methods not run under that protocol on that panel.

Method	Fixed origin					Walk-forward				
	OR	Carparts	Auto	RAF	M5	OR	Carparts	Auto	RAF	M5
AutoARIMA	13.8	27.1	8.6	50.0	10.4	24.6	34.4	8.2	52.3	—
AutoTheta	13.7	22.1	15.9	53.1	10.6	24.2	30.0	11.1	60.7	—
Tweedie-GLM	29.8	48.6	9.7	16.5	150.8	—	—	—	—	—
iETS	11.3	41.9	17.4	30.4	25.0	30.1	39.8	13.3	31.9	—
CP-Croston	75.3	24.8	11.0	24.0	4.2	60.9	26.7	9.7	23.1	13.7
CP-SBA	75.1	23.9	10.9	22.8	4.4	61.5	25.9	9.5	21.9	14.4
CP-TSB	32.8	17.7	17.8	42.3	15.8	61.1	22.9	13.7	43.3	20.5
CP-ADIDA	27.1	11.1	11.5	15.6	5.7	35.8	12.4	9.4	15.4	12.7
CP-IMAPA	27.5	10.8	11.6	16.3	5.9	38.1	11.9	9.5	16.6	12.1
ACI-ADIDA	—	—	—	—	—	6.8	9.5	8.5	15.1	3.0
TweedieGP	0.0	0.0	1.6	3.1	0.0	0.0	0.0	0.8	4.5	—
EBB	0.0	0.5	0.0	0.0	3.0	0.2	0.8	0.3	0.0	7.7
ACI-EBB	—	—	—	—	—	6.5	0.2	0.0	2.3	0.0
EBB (rule)	—	—	—	—	—	0.2	0.8	0.0	0.0	0.0

647-step external span (the original protocol forecasts 28 steps), a protocol difference rather than a method cost. Two caveats govern every timing in Table 28. First, all rows were measured on one machine in one session, and absolute values are not comparable across platforms. Second, Damato et al. [6] report 0.24 s per Carparts series on an M3 MacBook Pro; on our machine the released defaults give 1.89 s (single thread, median of twelve series) and the paper-stated budget of 100 iterations gives 0.97 s, because the released code doubles the iteration cap for Tweedie likelihoods on series with at most 500 observations.

F.8 Controlled-study diagnostics

In the length sweep used to isolate credibility leverage, the generator holds $K = 2$, 500 items per group, occurrence probabilities 0.05 and 0.80, log-size centers three units apart, and $w = 1$. The true partition’s pinball advantage falls monotonically from 1.53% at $T = 12$ (median 4.5 positives per item) to 0.01% at $T = 400$ (median 154), with NLL, Brier, and log-size MSE following the same direction. Across the main grid, the RMSE of $\hat{\tau}_g^2$ approximately doubles as w falls from 1.00 to 0.90 (0.037 $\rightarrow$ 0.144 at $T = 120$ and 0.074 $\rightarrow$ 0.149 at $T = 30$), directly measuring the resolution cost in Proposition 1.

The generator centers the groups symmetrically around a fixed grand mean so that increasing separation does not change total demand scale, and every arm is evaluated through the same hurdle-lognormal predictive representation under both likelihood-based proper scores and point metrics. A four-arm ladder—single pool, true labels with estimated hyperparameters, true labels with true hyperparameters, and the true conditional law—checks the implementation; the true conditional law is best on every metric in every cell.

Separation axis. The grid above pools the three separations $\{0.5, 1, 2\}$ of the original sweep, and the length sweep uses a single separation of three; the envelope of Figure 8 therefore mixes separations. To resolve them we rerun the same generator on a pre-registered grid: separation $\in \{0, 0.25, 0.5, 1, 2, 3\}$ log-units between adjacent group centers, $w \in \{1, 0.95, 0.90\}$, $T \in \{30, 120\}$, nine paired replicates per cell (seed 20260913), four groups of sixty items, $\tau^2 = 0.15$, $\sigma^2 = 1$, occurrence spread in logit scaled with the size separation as $0.8 \min(\text{sep}, 1)$ so that zero separation removes structure from both blocks (a first pass held it fixed at 0.8; it is archived and superseded). Each cell records the true partition’s paired pinball gain over a single pool, the gain of the paper’s mixture objective run on the same panel, the median size-block credibility under the single pool, and the share of item log-size variance explained by the true labels (R^2 , computed from the discounted item statistics so it includes sampling noise). Two remedies for the learned mixture that already exist in the released code are run as extra arms on the same panels: label sample-splitting (labels from one interleaved half, hyperparameters from the other) and credibility shrinkage of the group variance component, alone and combined. Table 30 summarizes the surface by separation and Table 31 reads it at the real panels’ coordinates. Real-panel R^2 uses the same statistic under the learned mixture labels at the selected discount and under the ADI/CV² taxonomy; we do

Table 19: Short-history scaling: initialization window truncated to its last L observations per series (evaluation span unchanged; SPL scale from the full window). Median credibility weights at the audited discount, mean SPL of EBB, TweedieGP, and CP-ADIDA, and EBB’s advantage over TweedieGP. Mixture labels are re-learned on the truncated window.

Panel	L	median n	$\lambda^{(o)}$	$\lambda^{(+)}$	EBB	TweedieGP	CP-ADIDA	Adv. (%)
Online Retail	14	14	0.823	0.405	0.8894	0.8920	1.0106	+0.3
	28	28	0.870	0.506	0.8862	0.8879	1.1519	+0.2
	56	56	0.893	0.546	0.8852	0.8859	1.1247	+0.1
	full	119	0.895	0.553	0.8849	0.8845	1.1239	-0.0
M5	28	28	0.910	0.812	1.6925	1.6626	1.7182	-1.8
	56	56	0.928	0.839	1.6645	1.6774	1.7082	+0.8
	112	112	0.930	0.844	1.6697	1.6558	1.6912	-0.8
	224	224	0.931	0.842	1.6637	1.5995	1.7092	-4.0
	448	448	0.931	0.842	1.6670	1.6063	1.7024	-3.8
	full	1294	0.931	0.841	1.6707	1.6228	1.7150	-3.0
Auto	6	6	0.278	0.867	0.3009	0.3221	0.3363	+6.6
	12	12	0.377	0.920	0.2948	0.3017	0.3317	+2.3
	full	18	0.421	0.940	0.2937	0.2985	0.3275	+1.6
Carparts	6	6	0.585	0.180	0.3199	0.3279	0.3858	+2.4
	12	12	0.560	0.231	0.3197	0.3173	0.3628	-0.8
	24	24	0.596	0.245	0.3206	0.3145	0.3585	-1.9
	full	45	0.628	0.224	0.3229	0.3214	0.3571	-0.5
RAF	6	6	0.037	0.000	0.2730	0.3079	0.4071	+11.3
	12	12	0.075	0.682	0.2696	0.2811	0.3506	+4.1
	24	24	0.135	0.790	0.2669	0.2788	0.3208	+4.3
	48	48	0.152	0.891	0.2674	0.2769	0.3141	+3.4
	full	72	0.106	0.917	0.2680	0.2763	0.3097	+3.0

Table 20: Mean SPL of alternative positive-size predictive laws computed from the same fitted EBB posterior (single fit, no bootstrap averaging), against TweedieGP.

Size law	OR	Carparts	Auto	RAF	M5
log-normal (released)	0.8849	0.3230	0.2937	0.2681	1.6730
log- t , $\nu=10$	0.8849	0.3228	0.2936	0.2680	1.6735
log- t , $\nu=5$	0.8850	0.3226	0.2940	0.2678	1.6744
Gamma (moment-matched)	0.8878	0.3227	0.2937	0.2673	1.6735
TweedieGP	0.8845	0.3214	0.2985	0.2763	1.6228

not use the learned $\hat{\tau}_g^2$ as a scale because it collapses to its floor under learned partitions (Section E.8). The simulation was run and summarized before the real panels were placed.

F.9 Released tables and artifacts

Complete per-dataset tables with RMSE, WRMSSE, all quantile levels, coverage, interval widths, fixed-corner ablations, and runtimes accompany the code release, together with the selection surfaces and per-run diagnostics behind every table in this paper.

F.10 Machine-readable significance output

The complete per-series results behind Table 26 and Table 23, including the RMSE tests and the Holm-adjusted families, accompany the code release as machine-readable CSV files.

F.11 MASE

For completeness we report MASE (per-series MAE scaled by in-sample naive MAE), the intermittent-demand literature’s traditional point metric. It inherits the absolute-error pathology quantified in Section E.5: the

Table 21: Item-specific forgetting rates with shrinkage strength κ selected on the outer validation tail. w^* is the panel-wide discount; “share $< w^*$ ” is the fraction of items that selected faster forgetting than the panel rate; $\kappa=\infty$ and $\kappa=0$ give the fully shrunk and unshrunk endpoints. External mean SPL, $B=20$.

Panel	w^*	Scalar	κ^*	Per-item	Δ (%)	share $< w^*$	$\kappa=\infty$	$\kappa=0$	TweedieGP
Online Retail	0.95	0.8849	10	0.8846	-0.03	0.19	0.8849	0.8845	0.8845
Carparts	0.9	0.3229	∞	0.3229	+0.00	0.00	0.3229	0.3298	0.3214
Auto	0.99	0.2937	30	0.2940	+0.10	0.14	0.2942	0.2931	0.2985
RAF	0.997	0.2680	100	0.2678	-0.07	0.16	0.2680	0.2675	0.2763
M5	0.95	1.6707	3	1.6515	-1.15	0.49	1.6751	1.8155	1.6228

Table 22: Separate occurrence discount $w^{(o)}$ with the size discount fixed at the selected w ; $w^{(o)*}$ is chosen on the initialization window’s validation tail. External mean SPL ($B=20$) for the tied and selected rates, the share of evaluation rows whose q90 forecast is exactly zero, and TweedieGP for reference.

Panel	w	$w^{(o)*}$	Tied	Selected	Δ (%)	q90 = 0 share	TweedieGP
Online Retail	0.95	tied	0.8849	0.8849	+0.00	0.362 $\rightarrow$ 0.362	0.8845
M5	0.95	0.98	1.6707	1.6683	-0.14	0.206 $\rightarrow$ 0.233	1.6228
Auto	0.99	tied	0.2937	0.2937	+0.00	0.000 $\rightarrow$ 0.000	0.2985
Carparts	0.9	0.8	0.3229	0.3204	-0.77	0.081 $\rightarrow$ 0.000	0.3214
RAF	0.997	tied	0.2680	0.2680	+0.00	0.279 $\rightarrow$ 0.279	0.2763

all-zero forecast attains the best MASE on Online Retail (1.84 vs. TSB 2.08, EBB 2.41), Carparts, and RAF, and zero-leaning methods (TSB, IMAPA) lead among real forecasters. On Auto, EBB (0.889) is second only to CrostonSBA (0.883). Full table in the repository (`mase_table.csv`). We treat MASE as descriptive, not decision-relevant, for the same reason as MAE.

F.12 Walk-forward artifacts

The complete point and probabilistic results are in Tables 12 and 4; released block-level artifacts retain the seven-day boundaries and every per-block prediction.

F.13 Neural baselines

Table 32 retains the earlier neural comparisons. Deep Renewal Processes (DRP) [27] are architecturally close to our occurrence-size decomposition, but the legacy subsample numbers are indicative only: we rebuilt the Python 3.8/MXNet 1.7 environment and launched the original code path, yet three configurations diverged to NaN between epochs 7 and 11 and failed during prediction. DeepAR and an RNN under prediction-only chaining reach lower MAE in one case but worse RMSE and RMSSE at both horizons. Neural decoding and a constant sufficient-statistic forecast are not timing-equivalent, so these rows should not support a broad efficiency ratio.

Two newly executed reference points provide a cleaner, narrower check. DeepState—the published Amazon ISSM successor available in GluonTS, not a reproduction of Seeger et al. [20]—obtains mean SPL 1.0652 on a 1,000-series Online Retail subset and 0.4608 on full Carparts. Chronos-Bolt-small obtains 0.8448 on an Online Retail 500-series subset, versus 0.8342 for EBB on exactly the same subset: a small 1.3% gap. Its Carparts result is only a reference point because the same-source Monash panel appears in the Chronos data repository and potential training contamination cannot be excluded. These subset results motivate an efficiency, determinism, and transparency claim, not a large foundation- model accuracy advantage.

F.14 Far-tail quantiles on the sparsest panels

Scaled pinball at $q \in \{0.95, 0.975, 0.99\}$ on the two panels where the standard levels discriminate weakly, under the identical fixed protocol (EBB raw, $B=20$; iETS not re-run at these levels). The zero forecast’s scaled pinball is *linear* in q (its pinball at level q is $q \mathbb{E}[y/\text{scale}]$), so the far tail is exactly where its apparent advantage must collapse; the question the table answers is who inherits the lead.

Three readings follow. First, **the saturation breaks as expected**: on RAF the zero forecast, marginally ahead

Table 23: Per-series mean SPL paired tests: EBB versus every comparator on each panel, Holm-adjusted within panel (paired t and Wilcoxon signed-rank; the family includes TweedieGP on every panel). Cells count comparators by outcome at the 5% level; named entries are the comparators for which EBB is *not* significantly better.

Panel	Family	Sig. better	Sig. worse	Undecided
Online Retail	10	9	0	TweedieGP
M5	10	5	1 (TweedieGP)	CP wrappers (4)
Auto	10	10	0	—
Carparts	10	9	0	TweedieGP
RAF	10	10	0	—

Table 24: Walk-forward per-series paired tests, Holm-adjusted within each (panel, reference) family; a comparison is significant when both the paired t and the Wilcoxon adjusted p are below 0.05. Families contain every walk-forward method with stored per-series quantiles; TweedieGP is not in the M5 family and the static conformal wrappers are not in the Online Retail family.

Panel	Reference	Family	Sig. better	Sig. worse	Undecided
Online Retail	EBB	7	6	0	TweedieGP
Carparts	EBB	11	9	0	ACI-EBB, TweedieGP
Auto	EBB	11	9	0	ACI-EBB, TweedieGP
RAF	EBB	11	11	0	—
M5	EBB	7	5	2 (ACI-ADIDA, ACI-EBB)	—
Online Retail	ACI-EBB	7	4	2 (EBB, TweedieGP)	ACI-ADIDA
Carparts	ACI-EBB	11	9	0	EBB, TweedieGP
Auto	ACI-EBB	11	10	0	TweedieGP
RAF	ACI-EBB	11	10	1 (EBB)	—
M5	ACI-EBB	7	7	0	—

at $q \leq 0.9$, falls behind every intensity-bearing method from $q=0.95$ on, and on Carparts it is last by a factor of two. Second, **tail leadership is model-specific**: official TweedieGP wins all three Carparts cells, while EBB is second in all three; on RAF, AutoARIMA wins q95, TweedieGP wins q97.5, and EBB wins q99 by a near tie. Third, this table deliberately narrows the claim: EBB’s most consistent advantage is aggregate SPL, not universal extreme-quantile dominance. The conformal wrappers still deteriorate in the RAF far tail, where their per-level calibration information is sparse.

F.15 Full-posterior inference on the same hierarchy

To separate the inference method from the model, we paired plug-in EB with a full-posterior sampler at each discount. Each pair uses the same single global pool, shared size-variance specification, discounted sufficient statistics, seed-controlled 500-series Online Retail subsample, evaluation span, and scoring code. The sampler uses Metropolis-within-Gibbs for the Beta hyperparameters, four deterministic chains of 4,000 sweeps, 2,000 burn-in iterations, and thinning by four. Both a stationary version ($w=1$) and a power-prior discounted version ($w=0.98$) are tested. The richer EBB arm runs its joint selection separately on this subsample and is reported only as context, not as an inference-method contrast.

Two readings. First, *the empirical-Bayes approximation is not a distributional-accuracy bottleneck*: paired full-posterior inference changes mean SPL by 0.007%/0.019%, q90 SPL by 0.018%/0.048%, and RMSSE by 0.025%/0.085% at $w = 1/0.98$. MAE is somewhat more sensitive but still changes only 0.70%/1.42%. Second, forgetting helps MAE under both inference methods, so that finding is not an artifact of the EB plug-in. The maximum split- $\hat{R}$ across $(\alpha, \beta, \mu_0, \tau^2, \sigma^2)$ is 1.0021, and the minimum effective sample size is 819 of 2,000 retained draws. What this experiment does *not* measure: the richer external state-space samplers [2, 20] remain unrun (Limitations, item 6); this conjugate single-pool Gibbs is far cheaper than those models and is a control for the inference method, not a stand-in for them. The plug-in EB path retains bit-reproducibility and requires no convergence diagnostics, but at this scale raw speed does not separate the two.

Table 25: Validation-tail rule for the calibration layer. On the 80/20 chronological split of the initialization window, EBB is initialized on the head (labels learned on the head only) and walked forward over the tail with and without the adaptive conformal layer; the layer is applied in deployment iff its tail scaled pinball ($q \in \{.5, .75, .9\}$) is lower. External columns are the walk-forward means of Table 4; “matches” marks panels where the rule picks the ex-post better variant.

Panel	Tail blocks	Tail plain	Tail ACI	Choice	Ext. EBB	Ext. ACI-EBB	Matches
Online Retail	4	0.7382	0.7967	plain	0.8540	0.9082	yes
M5	37	1.5686	1.4457	ACI	1.4851	1.3785	yes
Auto	3	0.3313	0.3304	ACI	0.2910	0.2901	yes
Carparts	9	0.8975	0.9061	plain	0.3145	0.3128	no
RAF	14	0.9846	1.0081	plain	0.2681	0.2743	yes

Table 26: Paired per-series MAE tests on Online Retail: EBB versus each alternative. Negative mean differences favor EBB; p -values are Holm-adjusted.

vs.	Mean diff	p_t	$p_{Wilcoxon}$
Stationary hierarchy	-0.165	$< 10^{-29}$	$< 10^{-11}$
TSB	-0.040	0.532	$< 10^{-155}$
TSB-tuned	-0.090	0.086	$< 10^{-72}$
ADIDA	-0.036	0.359	$< 10^{-88}$
IMAPA	-0.049	0.281	$< 10^{-97}$
CrostonSBA	-0.872	$< 10^{-26}$	$< 10^{-59}$
AutoARIMA	-0.891	$< 10^{-14}$	$< 10^{-12}$
AutoTheta	-2.644	$< 10^{-81}$	$< 10^{-104}$

F.16 Datasets

Online Retail: UCI 352; returns/zero-price removed, 99.5% quantity cap, daily aggregation densified with zeros. M5: Kaggle M5-Accuracy; seed-controlled 5,000-series sample. Both M5 protocols use a two-thirds initialization split; the learned mixture is selected at $w = 0.95$ on all three audited seeds. Auto/RAF: gluon-ts `intermittent-datasets` branch. Carparts: Zenodo 4656022, series without missing values and with at least one zero (2,509). Monthly panels use the last- h holdout of Damato et al. [6].

Table 27: Raw EBB interval quality across panels at nominal 80% coverage. No post-hoc calibration is used.

	OR	M5	Auto	Carparts	RAF
Coverage@80	0.892	0.846	0.907	0.931	0.921
AIW@80	9.25	2.81	9.21	1.47	0.94

Table 28: Instrumented wall-clock on Carparts, one machine, one session. “Sel.” marks totals that include the method’s own hyperparameter or candidate selection. AutoARIMA/AutoTheta share one StatsForecast call (split across the two); the five conformal wrappers share one residual-pool fit (split across the five). Rows below the rule are recorded from prior run metadata rather than re-timed: DeepState is a full-panel GluonTS/MXNet run, Chronos-Bolt is zero-shot inference on a 500-series subsample, and DRP is a 500-series Online Retail subsample (no Carparts run exists). All rows are CPU-only.

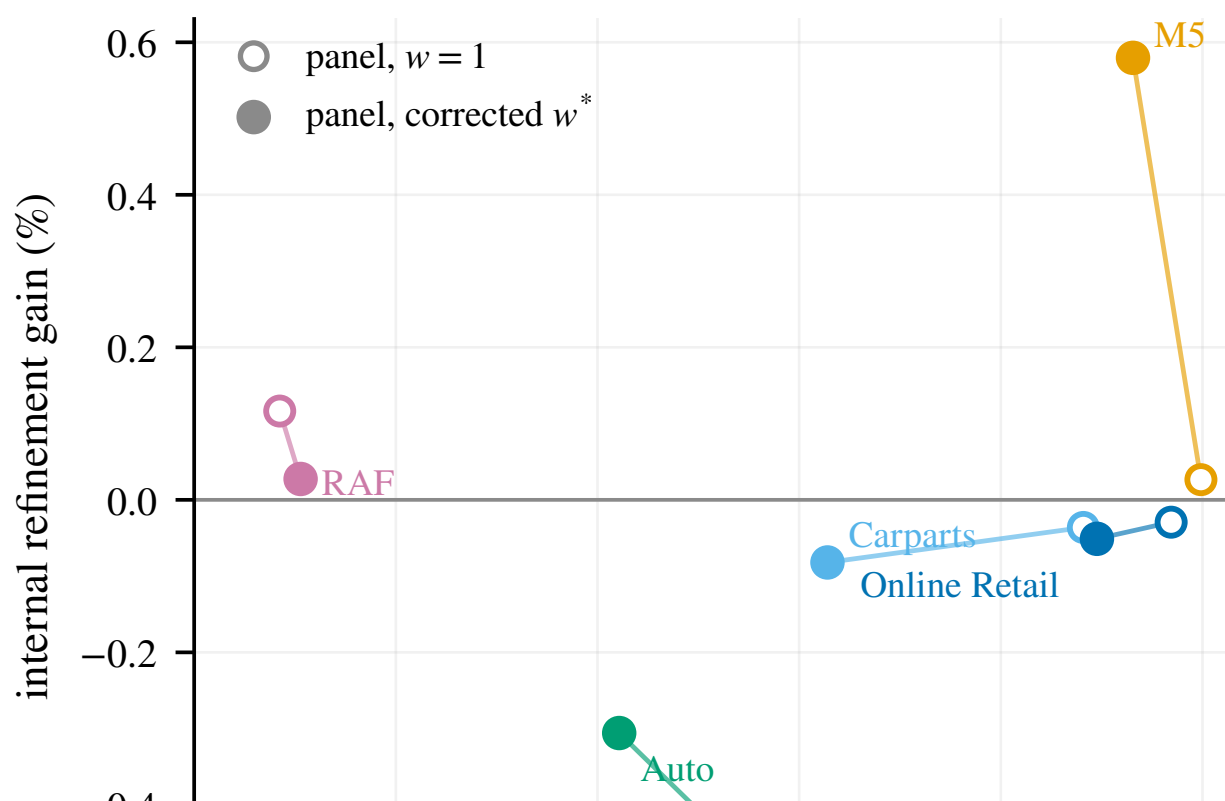
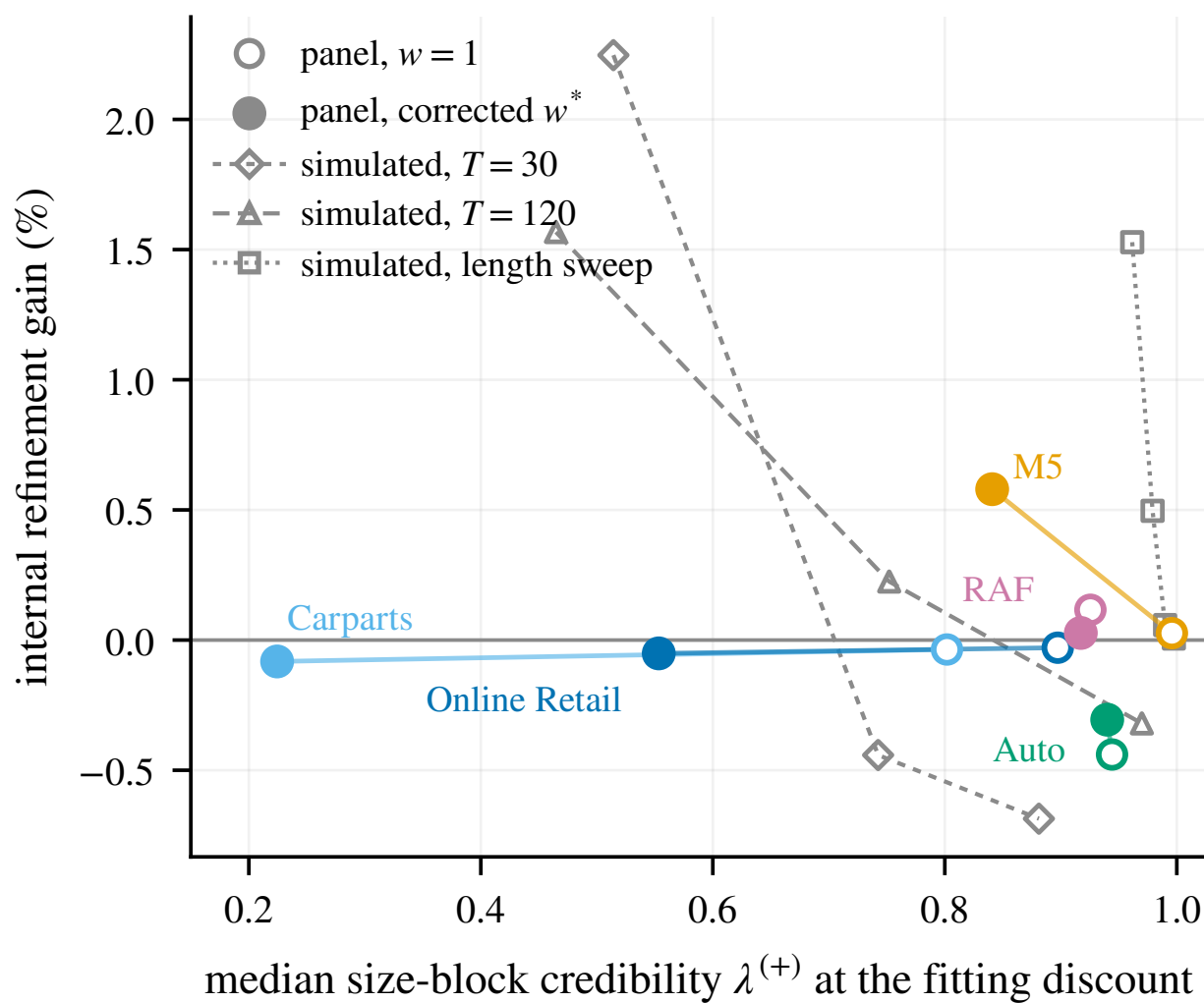
Method	Total (s)	s/series	Sel.	Runtime	Bit-repr.
EBB	46.7	0.019	yes	Python	Yes
TSB-tuned	3.8	0.002	yes	Python	No
CP wrappers (each)	13.3	0.005	no	Python	No
Tweedie-GLM	37.3	0.015	no	Python	No
iETS	55.3	0.022	no	R	No
AutoARIMA	56.2	0.022	no	Python	No
AutoTheta	56.2	0.022	no	Python	No
TweedieGP (12 workers)	302.1	0.120	no	Python/torch	No
DeepState	192.0	0.077	no	Python/MXNet	No
Chronos-Bolt-small	8.3	0.017	no	Python/torch	No
DRP (Deep Renewal)	1200.0	2.400	no	Python/MXNet	No

Table 29: Walk-forward wall-clock per method and panel (seconds unless marked in hours), covering every block of the protocol of Table 4: initialization, all per-block refits or updates, and prediction. StatsForecast, conformal, and ACI rows run single-process; TweedieGP runs its released code with twelve worker processes; EBB runs on one core and includes its initialization. AutoARIMA and AutoTheta share one call. [†]Projected from the measured 7,618s full fit on the 5,000-series M5 sample at the Online Retail cadence of one refit per four blocks (24 refits); at one refit per block the projection is about 197 h. Neither was run.

Method	Online Retail	Carparts	Auto	RAF	M5
AutoARIMA	1169	1.1 h	763	2.7 h	—
AutoTheta	1169	1.1 h	763	2.7 h	—
iETS	1795	270	213	1254	—
CP-ADIDA	—	9.6	4.4	38.0	465
ACI-ADIDA	126	7.4	8.9	27.3	581
TweedieGP	1.7 h	2686	2139	2.4 h	≈ 51 h [†]
EBB	22.4	3.4	3.6	13.5	36.6
ACI-EBB	52.3	4.2	4.8	13.3	217
EBB (rule)	22.4	3.4	4.8	13.5	217

Table 30: Separation-resolved controlled study, averaged over the six (w, T) cells at each separation: true-label R^2 , the true partition’s gain over a single pool, the learned mixture’s gain, the same with label sample-splitting and credibility shrinkage combined, and the median number of components the mixture selects.

Separation	R^2	True (%)	Learned (%)	Learned+fix (%)	K
0	0.01	+0.07	-0.02	-0.36	1
0.25	0.19	+0.12	-0.01	-0.30	1
0.5	0.47	+0.70	-0.92	-0.94	2
1	0.75	+0.20	-1.47	-0.86	2
2	0.93	-0.08	-0.71	-0.37	3
3	0.97	-0.30	-0.37	-0.51	4



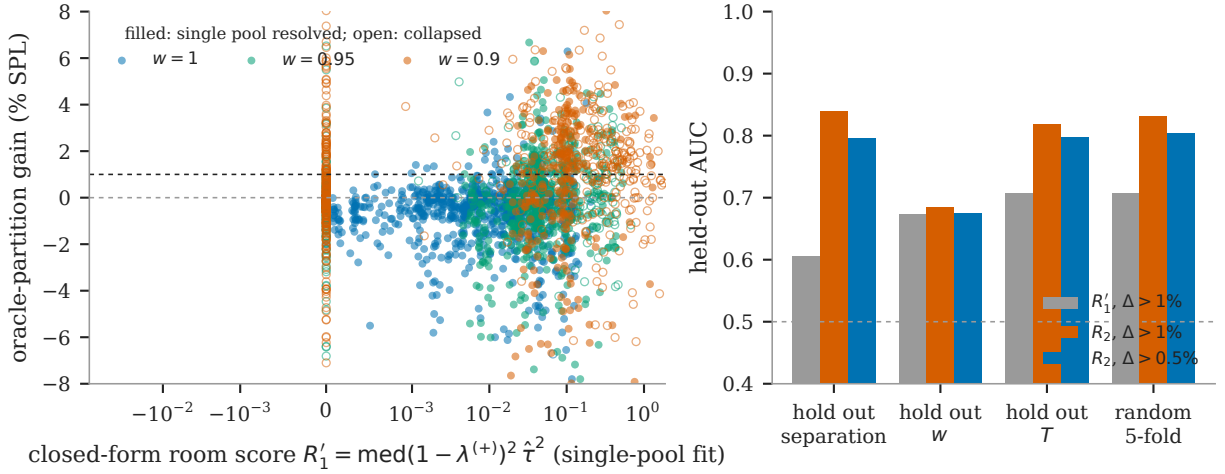


Figure 9: **Screening for pooling room before any partition is fitted.** Left: the closed-form score computed from the single-pool fit against the oracle partition’s realized gain on 2,700 simulated panels, colored by discount; open markers are panels whose single-pool variance component collapsed. Right: held-out AUC for the event that the oracle gain exceeds one percent (or half a percent), when whole separation levels, discounts, or lengths are held out, or under random folds; R'_1 is the closed-form score and R_2 a logistic screen on the same fitting-window quantities.

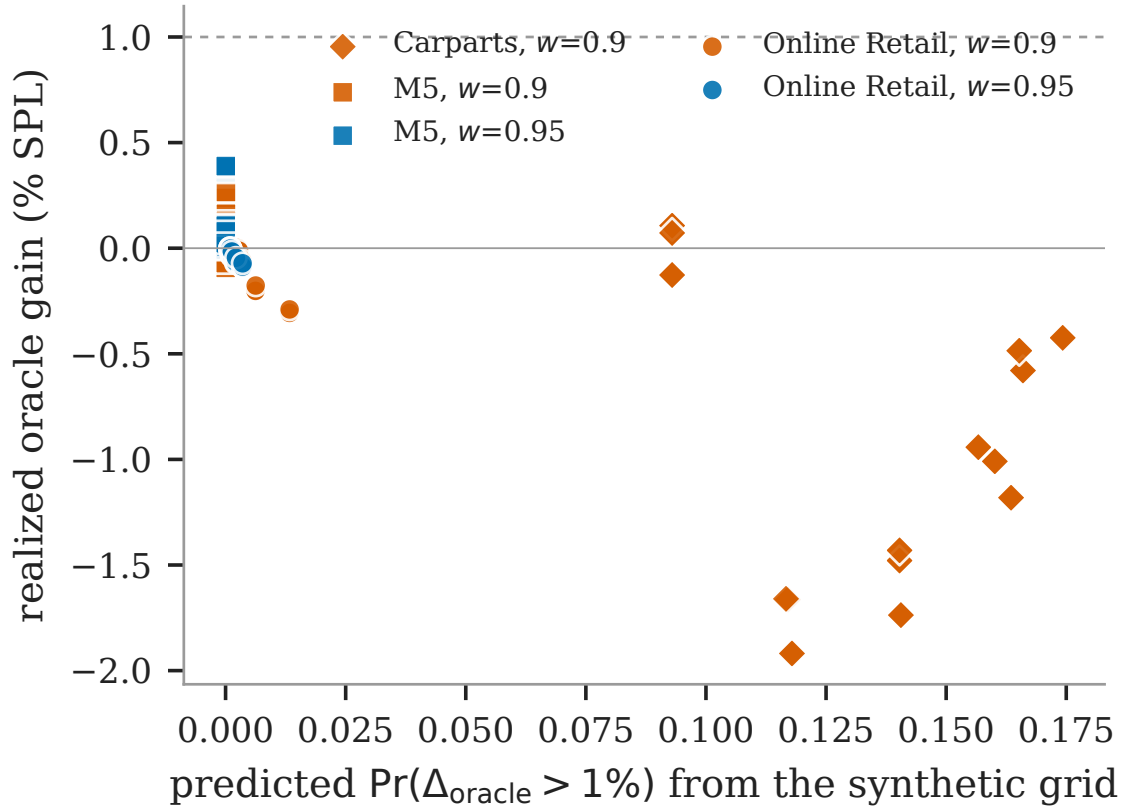


Figure 10: Real-series transfer of the screen: predicted probability of material room (from the simulated grid) against the realized oracle gain of the injected partition, by panel and discount.

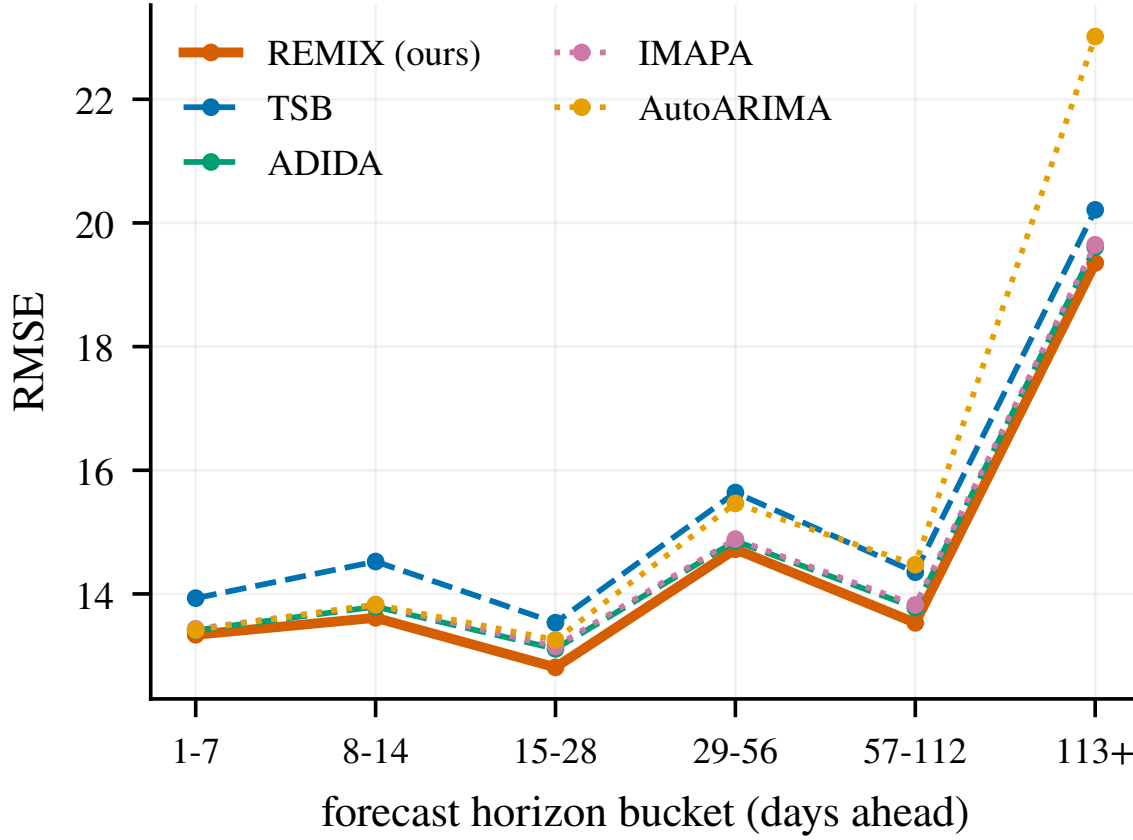


Figure 11: Online Retail RMSE by fixed-origin forecast-horizon bucket. EBB is lowest in every bucket, so the aggregate result is not created by averaging only over distant horizons. MAE differences among the leading methods are within 1–2% by bucket.

Table 31: Real panels placed on the separation-resolved surface. w is the simulated discount matching the panel’s selected one; R^2 is the size-block separation under the learned partition and under the taxonomy, and “family R^2 ” the true-label R^2 of the simulated family at the panel’s $\lambda^{(+)}$ ($T=120$). “Room” is the true partition’s gain interpolated along $\lambda^{(+)}$ within the matching w at each length; “sim. learned” the learned mixture’s gain at the same point ($T=120$); “realized” the panel’s internal refinement gain (Figure 8).

Panel	w	$\lambda^{(+)}$	R^2 learned	R^2 tax.	family R^2	Room $T=30$	Room $T=120$	Sim. learned	Realized
Online Retail	0.95	0.553	0.44	0.17	0.63	-0.52	+0.63	-0.87	-0.05
M5	0.95	0.841	0.79	0.30	0.86	-0.34	-0.45	-1.21	+0.58
Auto	1	0.940	0.82	0.14	0.72	-0.50	-0.33	-0.45	-0.31
Carparts	0.9	0.224	0.68	0.01	0.55	+2.01	+1.73	-0.75	-0.08
RAF	1	0.917	0.95	0.09	0.62	-0.48	-0.41	-0.58	+0.03

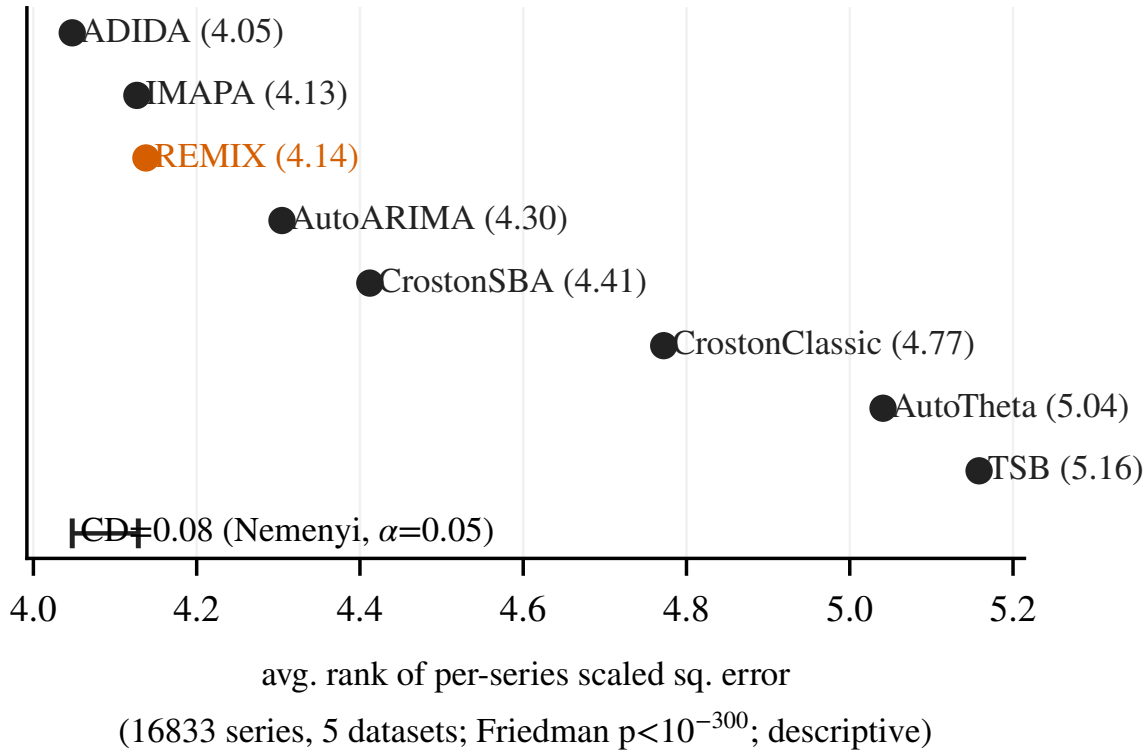


Figure 12: Average ranks of per-series scaled squared errors over the 16,833 series with complete saved row-level point predictions across the five panels. The critical-difference bar is shown only for scale: series, rather than datasets, are the blocks, so this is descriptive and not a valid Friedman–Nemenyi test.

Table 32: Neural comparisons on Online Retail. Top: probabilistic (unscaled pinball, 500-series subsample, fixed protocol). Bottom: point metrics under prediction-only chaining at two horizons.

	q10	q25	q50	q75	q90	Time
DeepRenewal	0.358	0.895	1.790	2.684	2.941	~20 min
Hierarchical EB	0.387	0.914	1.772	2.390	2.186	1 s
	MAE		RMSE		RMSSE	
	$h=10$	$h=100$	$h=10$	$h=100$	$h=10$	$h=100$
RNN	3.257	3.132	11.938	12.538	0.875	8.197
DeepAR	6.245	7.288	17.221	18.623	7.151	9.986
Hierarchical EB	4.639	4.495	11.284	11.733	0.855	8.170

Model	Carparts			RAF		
	q95	q97.5	q99	q95	q97.5	q99
CP-Croston	0.413	0.290	0.168	0.488	0.537	0.526
CP-SBA	0.412	0.290	0.168	0.489	0.542	0.529
CP-TSB	0.374	0.263	0.160	0.506	0.601	0.552
CP-ADIDA	0.384	0.275	0.159	0.493	0.579	0.557
CP-IMAPA	0.378	0.270	0.156	0.493	0.581	0.564
AutoARIMA	0.385	0.332	0.301	0.460	0.395	0.356
AutoTheta	0.367	0.316	0.285	0.464	0.399	0.359
Tweedie-GLM	0.581	0.445	0.305	0.497	0.439	0.367
TweedieGP	0.355	0.222	0.128	0.462	0.352	<u>0.247</u>
Zero	0.710	0.729	0.740	0.507	0.520	0.529
EBB	<u>0.365</u>	<u>0.251</u>	<u>0.144</u>	<u>0.461</u>	<u>0.364</u>	0.246

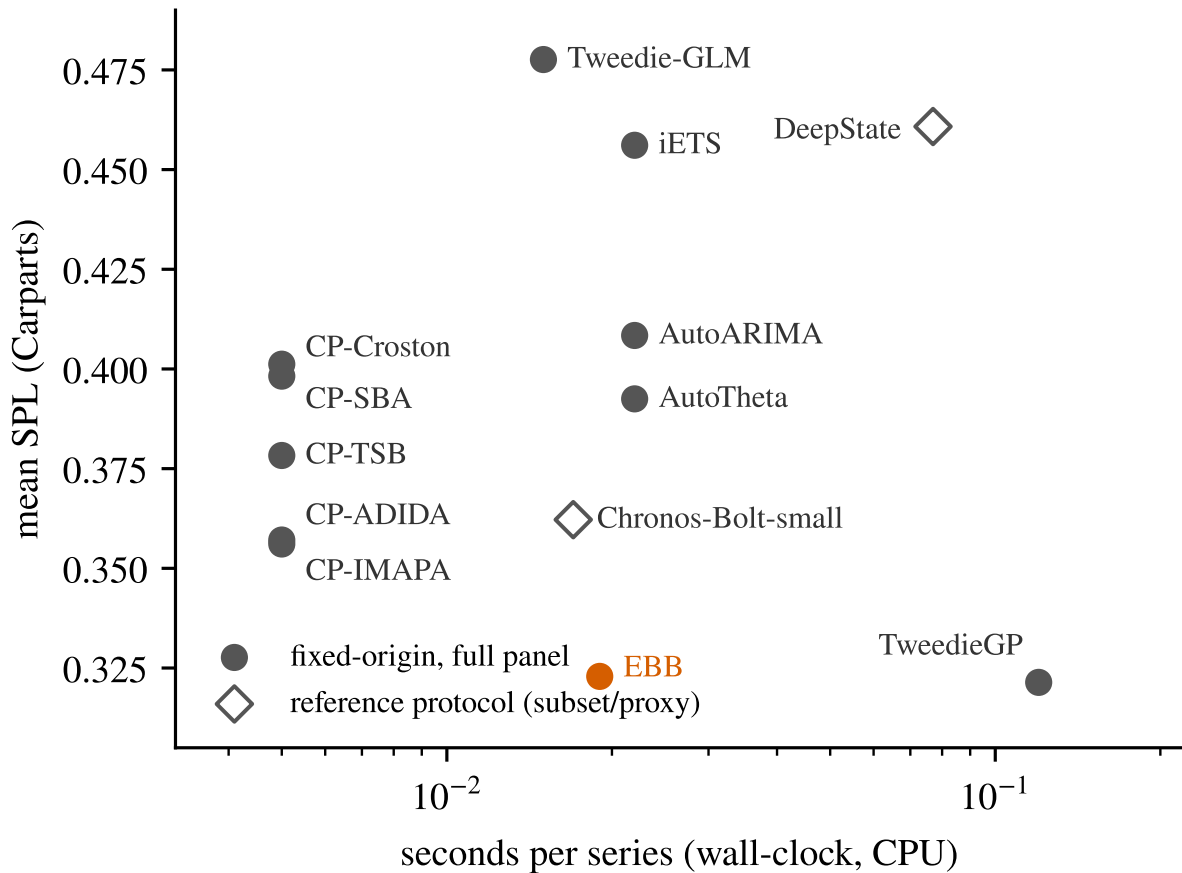


Figure 13: Wall-clock cost per series against mean SPL on Carparts. Filled circles are fixed-origin full-panel runs from Table 28; open diamonds are the reference points with subset or proxy protocols. All methods run on CPU.

	SPL mean	SPL q90	MAE	RMSSE	sec.
EBB (incl. selection)	0.7942	1.3682	5.6259	4.6712	86.4
EB global, $w=1$	0.7919	1.3588	5.5504	4.6684	0.2
Gibbs global, $w=1$	0.7919	1.3590	5.5890	4.6696	2.7
EB global, $w=0.98$	0.7935	1.3658	5.4774	4.6694	0.2
Gibbs global, $w=0.98$	0.7937	1.3665	5.5549	4.6734	2.7